Chapter 01 – MOS Transistor

Winter Term 2025/26

Prof. Dr.-Ing. Jens Anders

Overview

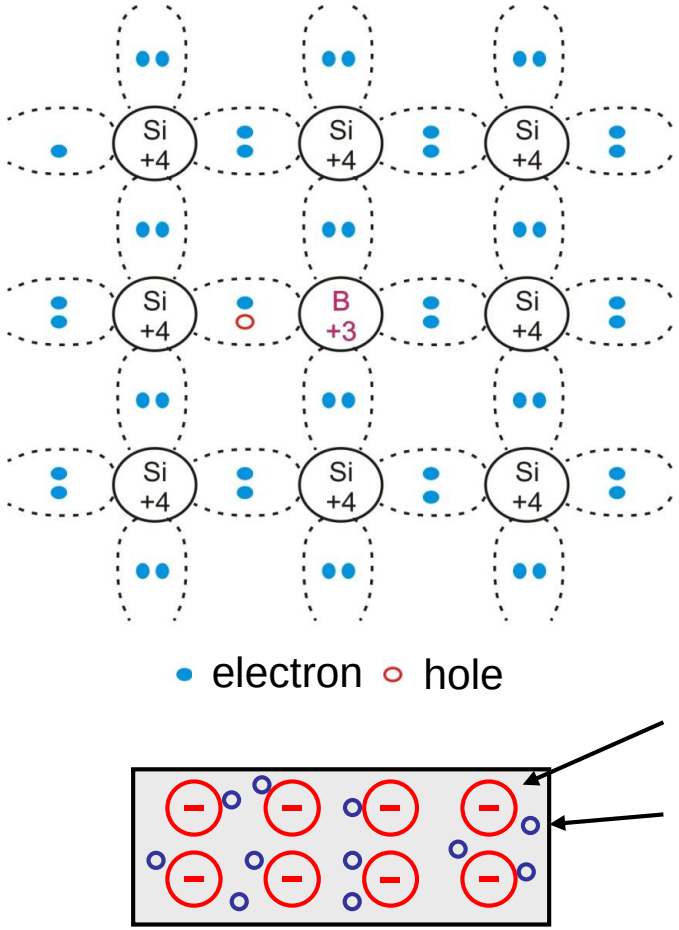
- MOS Transistor
  - $i(v)$ – Characteristics
  - Second-order Effects
  - Short-channel Effects
- Passive Devices & Wiring

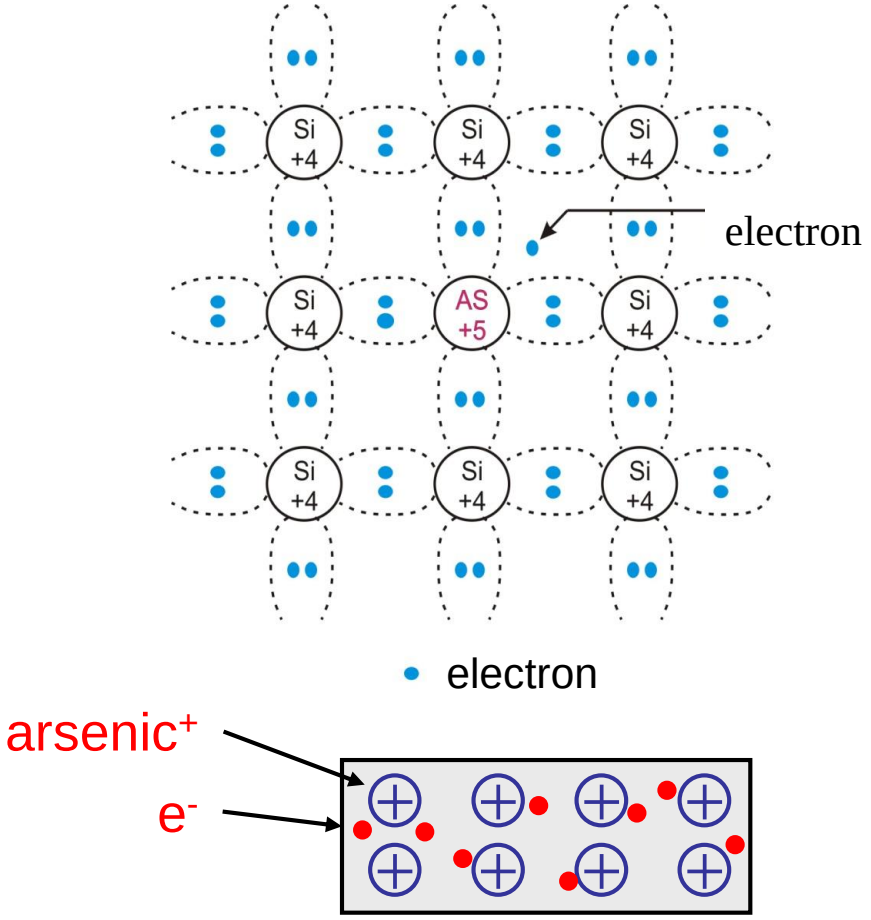
Overview

- **MOS Transistor**
  - **$i(v)$ – Characteristics**
    - **Configuration / Biasing**
    - Operating Regions
    - NMOS vs. PMOS
  - Second-order Effects
  - Short-channel Effects
- Passive Devices & Wiring

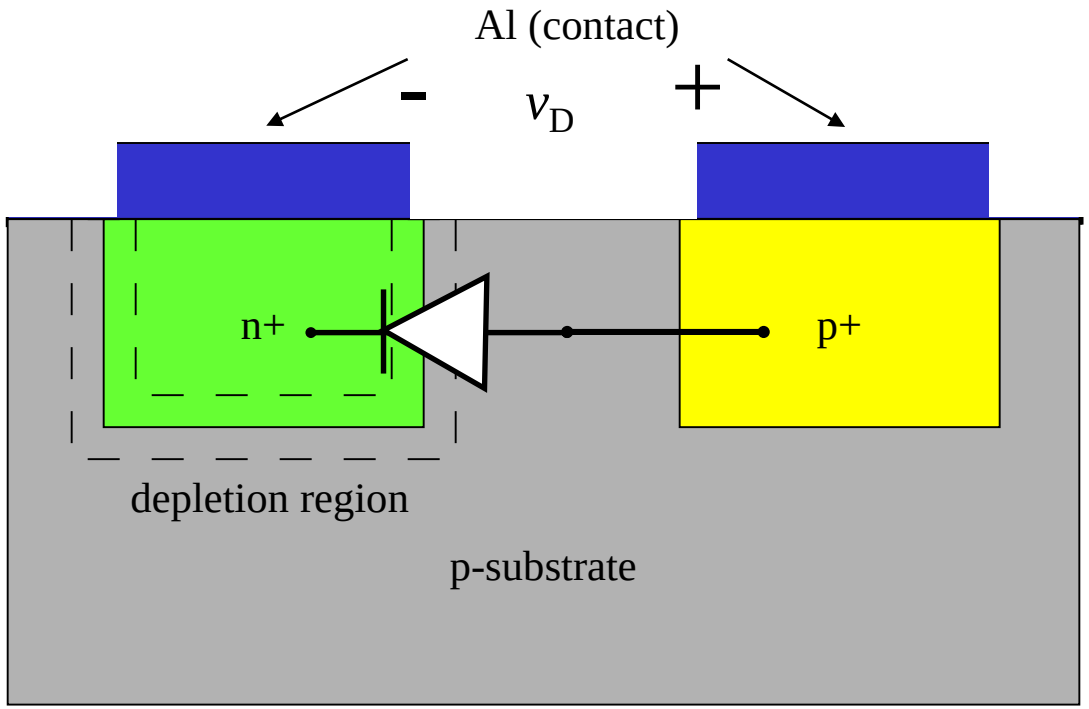
Semiconductor – Silicon – Doping

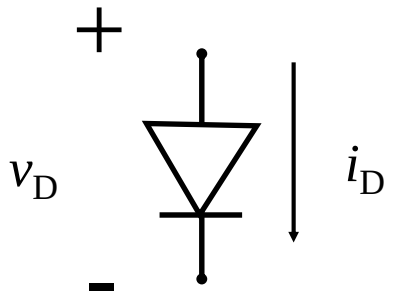
Adapt the electrical properties of silicon by doping the semiconductor with group 3 elements (acceptors: boron – B) or group 5 elements (donors: arsenic – As).





Diode



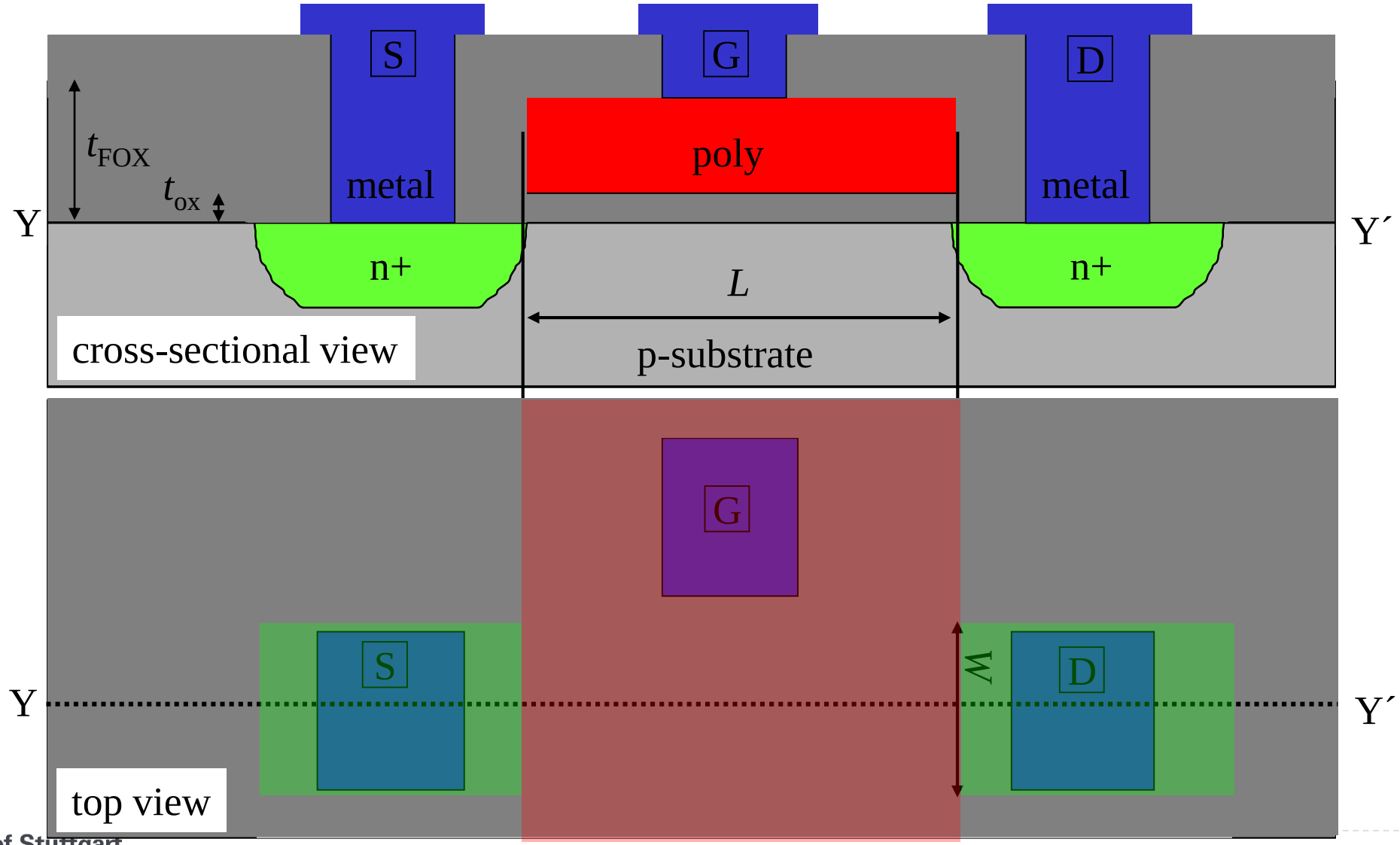


$i_D(v_D)$ – characteristic

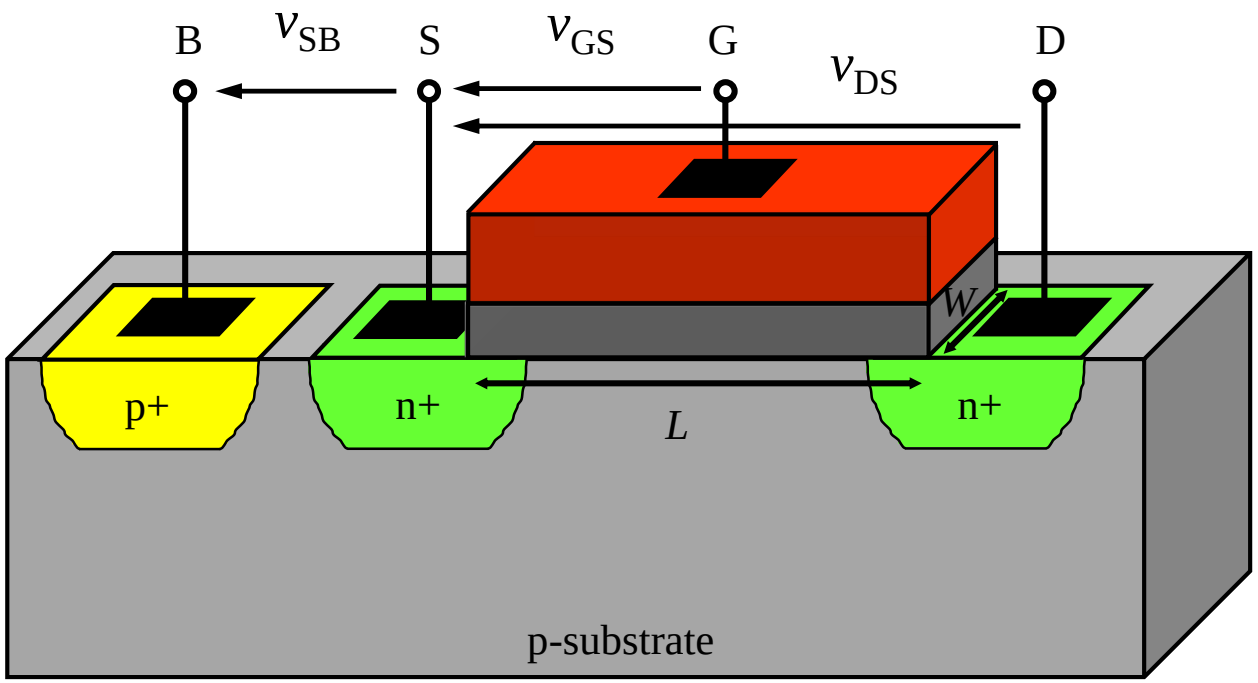
$$i_D = i_S\left(e^{v_D/V_T} - 1\right)$$

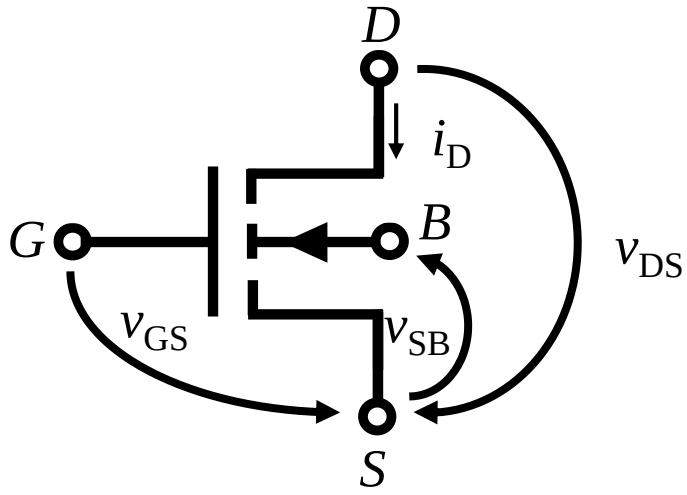
Parasitic forward-biased diodes are very critical in MOS technology!
--> parasitic BJT, latch-up, etc.

MOS-Transistor



Bird's View – MOS Symbol & Terminals





exemplary values:

$L = 0.12\ \mu m$

$W = 0.24\ \mu m$

$t_{ox} = 2.7\ nm$

Process Development

supply voltage	process (minimal L)	thickness of gate oxide
5 V	0.5 µm	8 nm
3.3 V	0.35 µm	5.4 nm
2.5 V	0.25 µm	3.2 nm
1.8 V	0.18 µm	2.0 nm
1.2 V	0.13 µm	1.5 nm
1.0 – 1.2 V	90 nm	1.2 nm
1.0 V	65 nm	1.2 nm
1.0 V	45 nm	2.5 nm (SiO_2: 1.0 nm)
1.0 V	32 nm	2.3 nm (SiO_2: 0.9 nm)

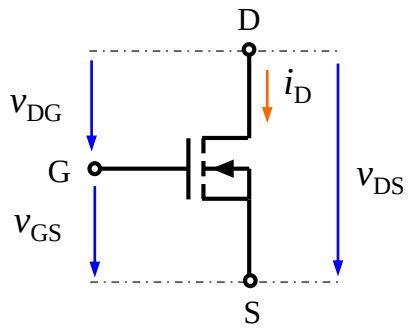
SiO_2 replaced by new materials with higher relative dielectric constant ("high k" materials, e.g., Zirconium dioxide,ZrO_2 or, Hafnium dioxide, HfO_2)
in order to continue the shrinking process.

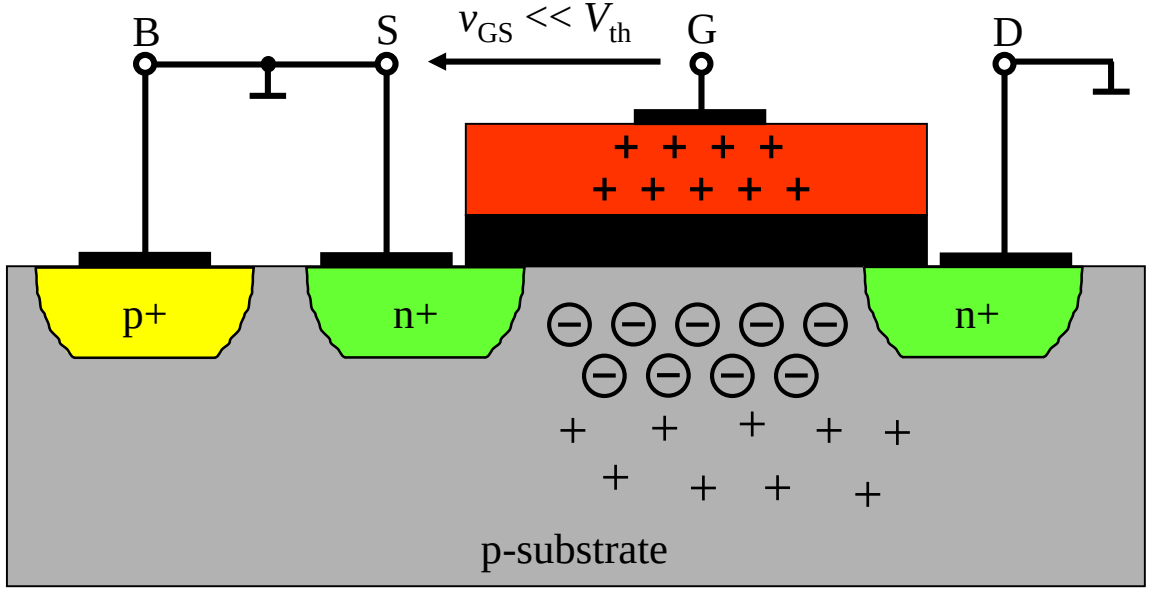
$$\varepsilon_{r,SiO_2} = 3.9 \leftrightarrow \varepsilon_{r,HfO_2} = 25$$

Overview

- **MOS Transistor**
  - **$i(v)$ – Characteristics**
    - Configuration / Biasing
    - **Operating Regions**
    - NMOS vs. PMOS
  - Second-order Effects
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$i(v)$ - Characteristic – Subthreshold Region





Depletion region is formed!

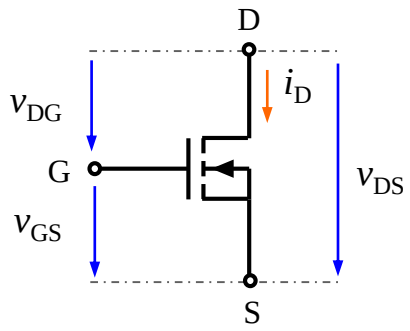
However, the transistor is still turned off!
There is no inversion layer yet!

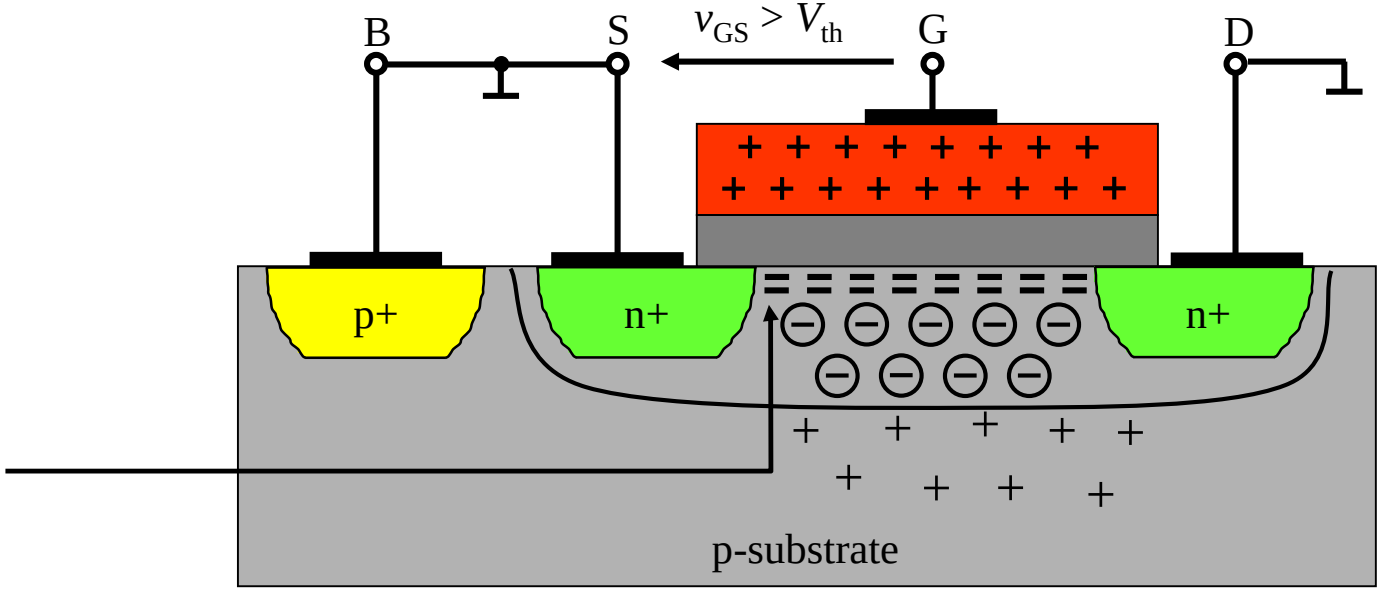
<u>subthreshold region</u>

$v_{GS} \ll V_{th}$ V_{th} – threshold voltage

$$i_D \approx 0\ A$$

$i(v)$ - Characteristic – Strong Inversion





charge in inversion layer

(gate-channel plate capacitor)

$$Q_{\mathrm{i}} = -C'_{\mathrm{ox}} W L (v_{\mathrm{GS}} - V_{\mathrm{th}})$$

with Oxide capacitor per unit

area, $C'_{\mathrm{ox}} = \frac{\varepsilon_{\mathrm{ox}}}{t_{\mathrm{ox}}}$

<u>(strong) inversion</u>

$$v_{\mathrm{GS}} > V_{\mathrm{th}}$$

Inversion layer is established, current may flow.

$i(v)$ - Characteristic – Strong Inversion

- charge in inversion layer

$$Q_\mathrm{i} = -C'_\mathrm{ox} W L (v_\mathrm{GS} - V_\mathrm{th})$$

- current flows if $v_\mathrm{DS} > 0$ V, electrons move with drift velocity:

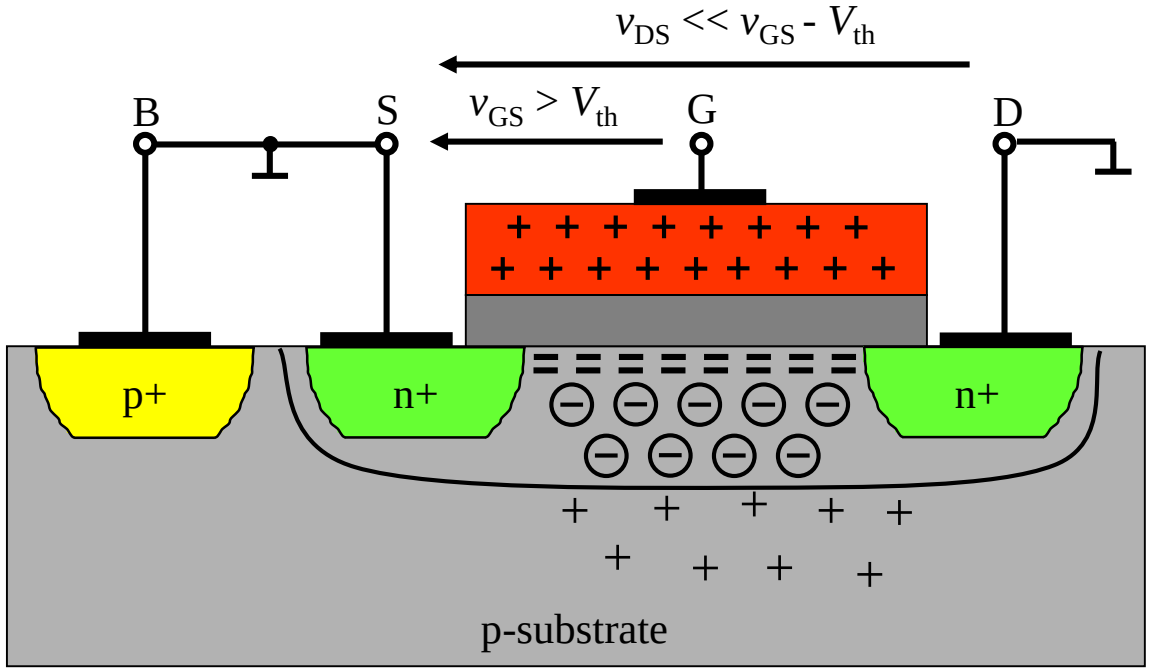
$$\vec{v}_\mathrm{drift} = -\mu_\mathrm{n} \vec{E} = -\mu_\mathrm{n} \frac{\vec{v}_\mathrm{DS}}{L}$$

- passed travel time until total channel charge has reached the drain

$$|t_\mathrm{tr}| = \frac{L}{|\vec{v}_\mathrm{drift}|} = \frac{L^2}{\mu_\mathrm{n} |\vec{v}_\mathrm{DS}|} = \frac{L^2}{\mu_\mathrm{n} v_\mathrm{DS}}$$

- for $v_\mathrm{DS} \ll$, current results from $-Q_\mathrm{i} = i_\mathrm{D} t_\mathrm{tr}$

Distinguish two operating regions: linear and saturation!



voltage responsible for

- charge
- current flow

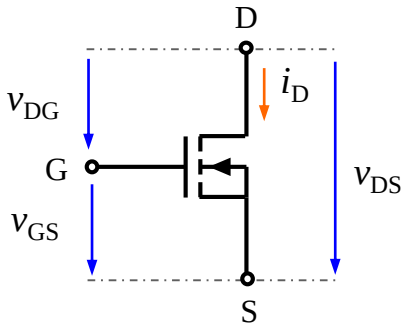
$$i_\mathrm{D} = \mu_\mathrm{n} C'_\mathrm{ox} \frac{W}{L} (v_\mathrm{GS} - V_\mathrm{th}) v_\mathrm{DS}$$

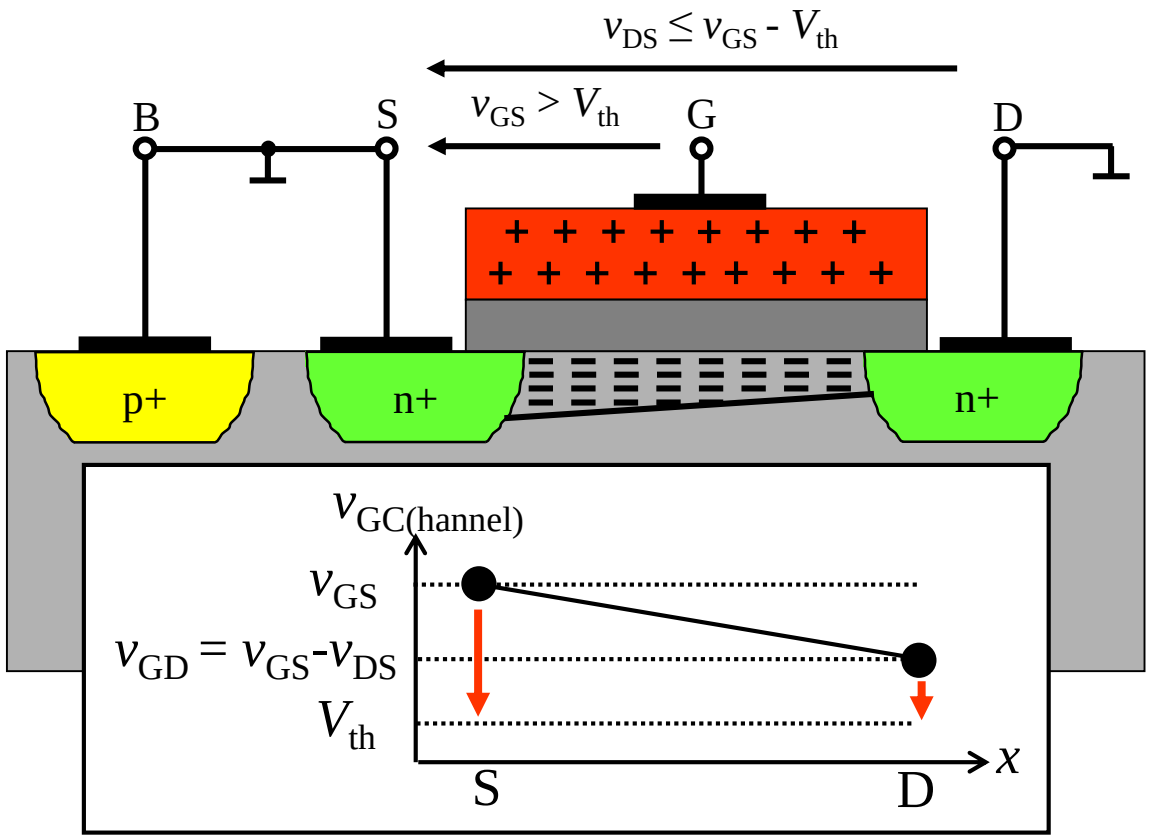
$i(v)$ - Characteristic – Linear Region

linear region

$v_{GS} > V_{th}$

$v_{DS} \leq v_{GS} - V_{th}$





charge density in channel at point X with infinitesimal width Δx

$$\Delta Q_i(x) = -C'_{ox} W \Delta x (v_{GC}(x) - V_{th})$$

$$i_D = \beta \left(v_{GS} - V_{th} - \frac{v_{DS}}{2} \right) v_{DS}$$
$$= \beta_{0n} \frac{W}{L} \left(v_{GS} - V_{th} - \frac{v_{DS}}{2} \right) v_{DS}$$

Derivation will be presented in our concentration course *Analog CMOS Circuit Design*.

with $\beta_{0n} = \mu_n C'_{ox} = \mu_n \frac{\varepsilon_{ox}}{t_{ox}}$

MOS transistor in linear region = voltage-controlled resistor → $f(\frac{W}{L}, v_{GS})$.

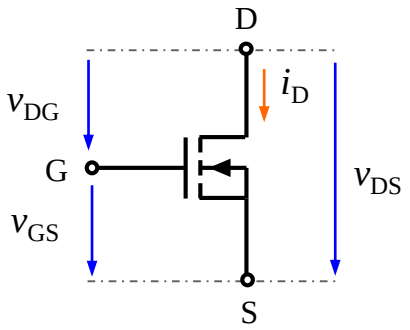
$i(v)$ - Characteristic – Saturation Region

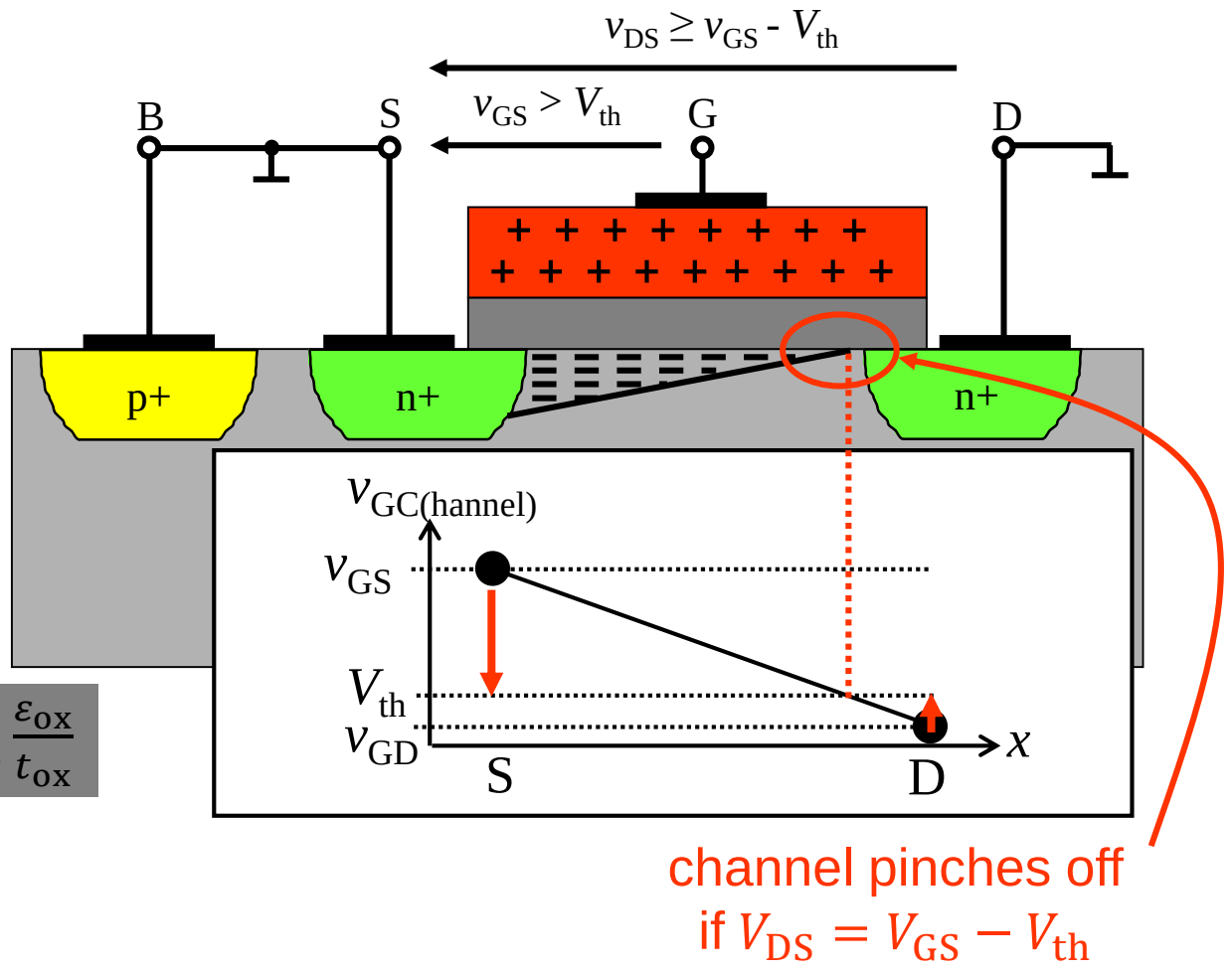
saturation region

$v_{GS} > V_{th}$

$v_{DS} \geq v_{GS} - V_{th}$

$v_{GC} < V_{th} \rightarrow$ no inversion





$$i_D = \frac{\beta}{2}(v_{GS} - V_{th})^2 = \frac{\beta_{0n}}{2}\frac{W}{L}v_{GSeff}^2$$

with $\beta_{0n} = \mu_n C'_{ox} = \mu_n \frac{\varepsilon_{ox}}{t_{ox}}$

channel pinches off if $V_{DS} = V_{GS} - V_{th}$

i_D is no longer a function of v_{DS}!

MOS transistor in saturation = voltage-controlled current source $\rightarrow f(\frac{W}{L}, v_{GS})$.

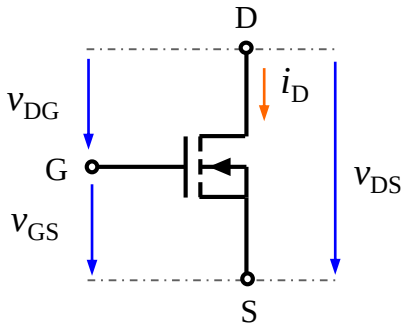
$i(v)$ - Characteristic – Saturation Region

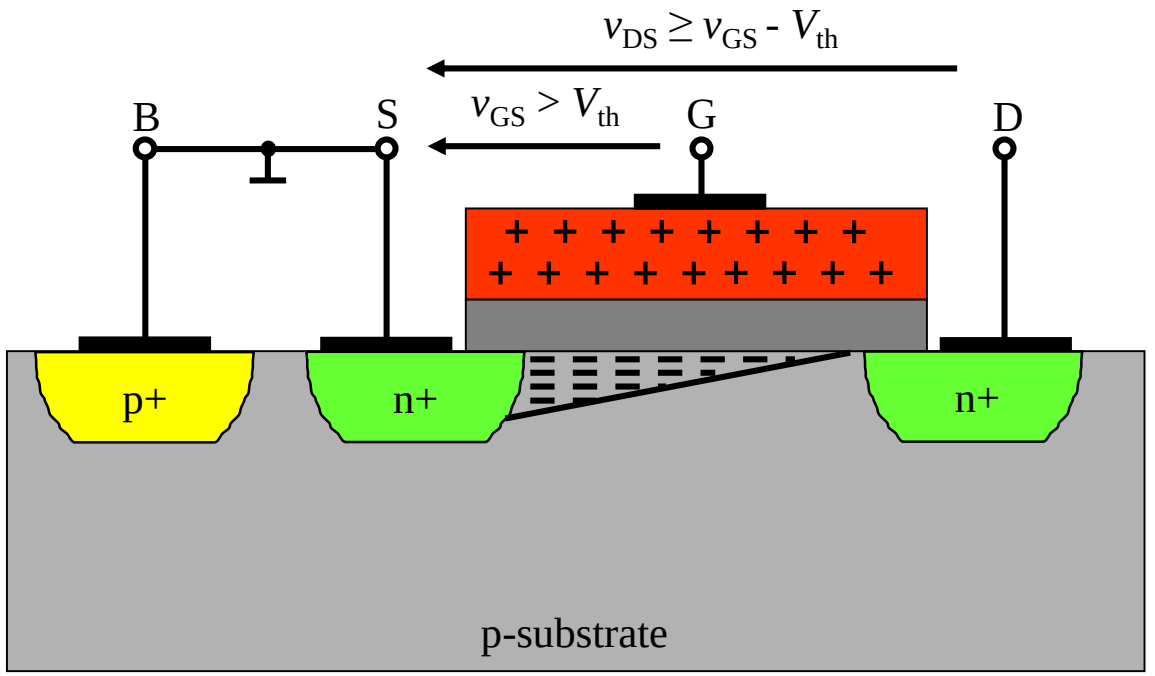
saturation region

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$$i_D = \frac{\beta}{2}(v_{GS} - V_{th})^2 = \frac{\beta_{0n}}{2}\frac{W}{L}v_{GSeff}^2$$

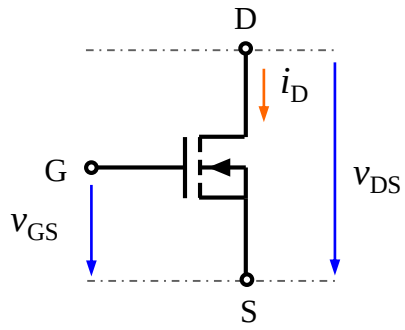
with $\beta_{0n} = \mu_n C'_{ox} = \mu_n \frac{\varepsilon_{ox}}{t_{ox}}$

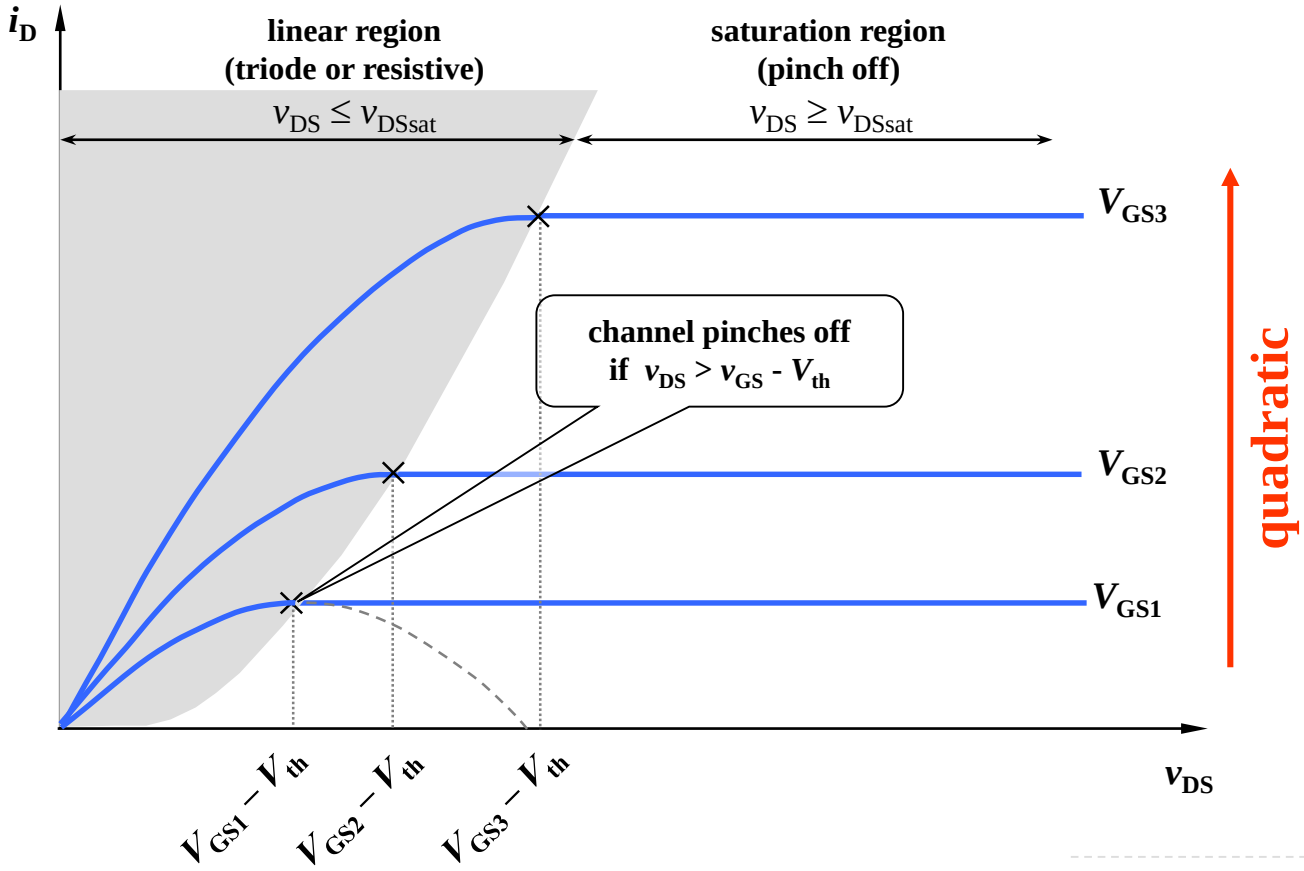
V_{GS} (mV)	V_{th} (mV)	$V_{GS} - V_{th}$ (mV)	v_{DS} (mV)	i_D
300	200	100	>100	I_D
400	200	200	>200	$4I_D$

Ideal $i_D(v_{DS})$ - Transfer Characteristic

$$i_D = \beta_{0n} \frac{W}{L} \left(v_{GS} - V_{th} - \frac{v_{DS}}{2} \right) v_{DS}$$

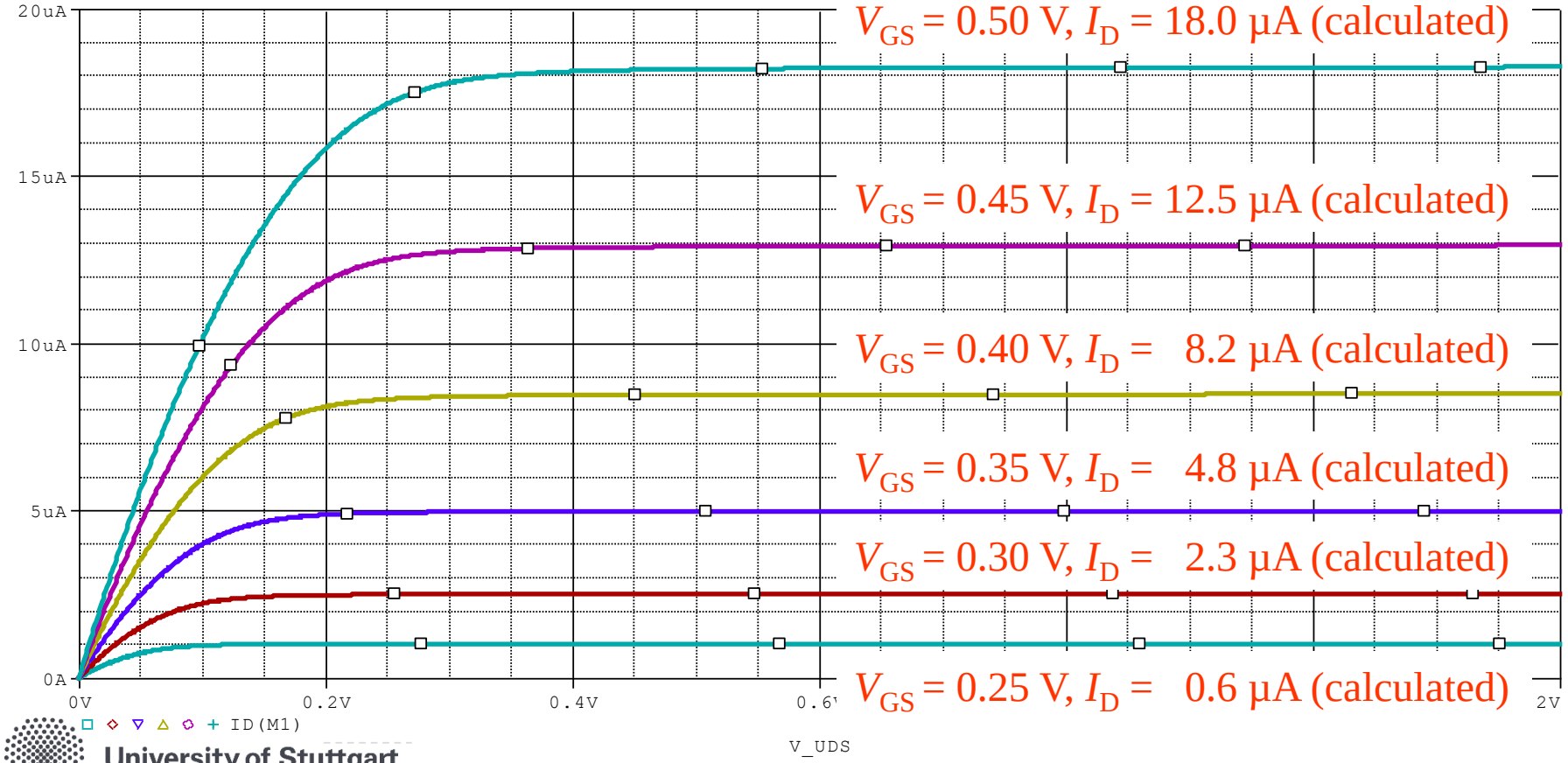
$$i_D = \frac{\beta_{0n}}{2} \frac{W}{L} (v_{GS} - V_{th})^2$$

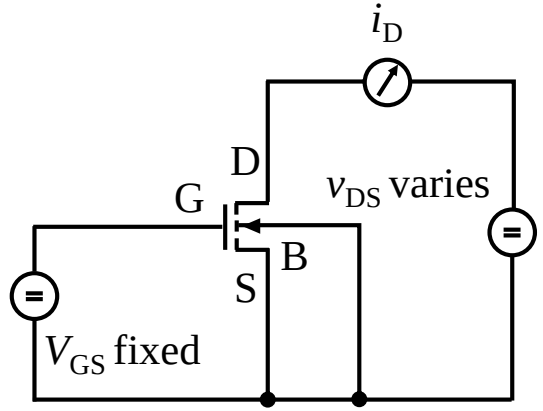




Case Study – $i_D(v_{DS})$ - Transfer Characteristic (CMOS 0.12 µm)

- 0.12 µm CMOS technology, $V_{thn} = 0.19\,\text{V}, \beta_{0n} = 370\,\mu\text{A/V}^2$
- Simulate i_D versus v_{DS} of a $\frac{W}{L} = \frac{100}{100}$ device for different V_{GS}!
- Compare the results with the calculation!





$$i_D = \frac{\beta_{0n}}{2}\frac{W}{L}(v_{GS} - V_{th})^2$$

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- **MOS Transistor**
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pMOS Transistor – Strong Inversion

Everything is inverted!

p-substrate → n-substrate

p+ doping (bulk) → n+ doping

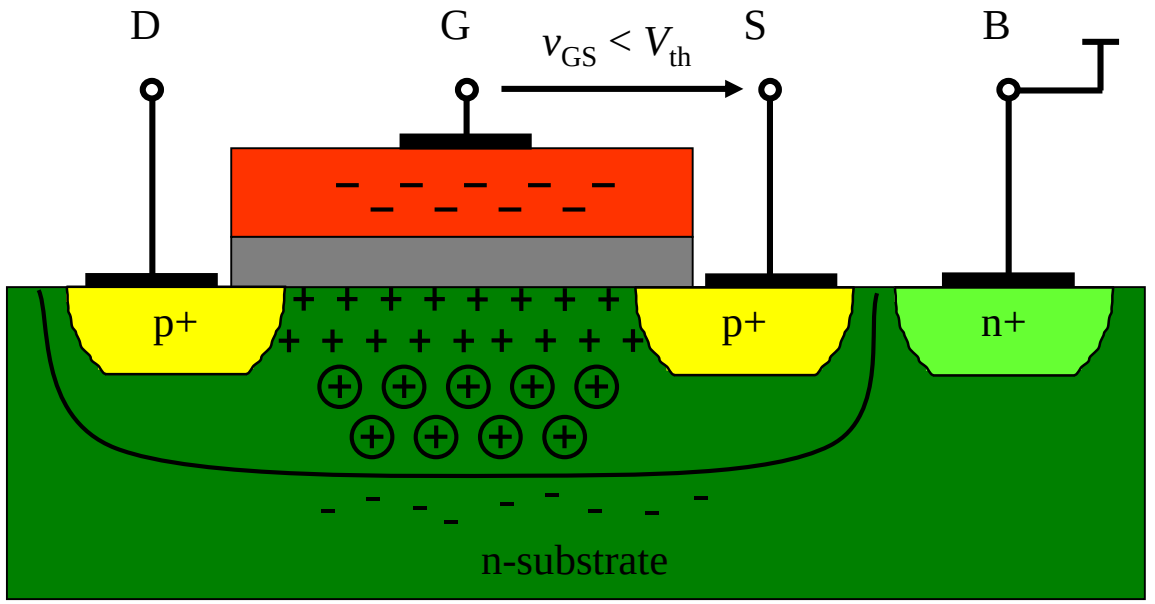
n+ doping (source, drain) → p+ doping

negative V_{th}

<u>(strong) inversion</u>

$$V_{\mathrm{th}} < 0\ \mathrm{V}$$

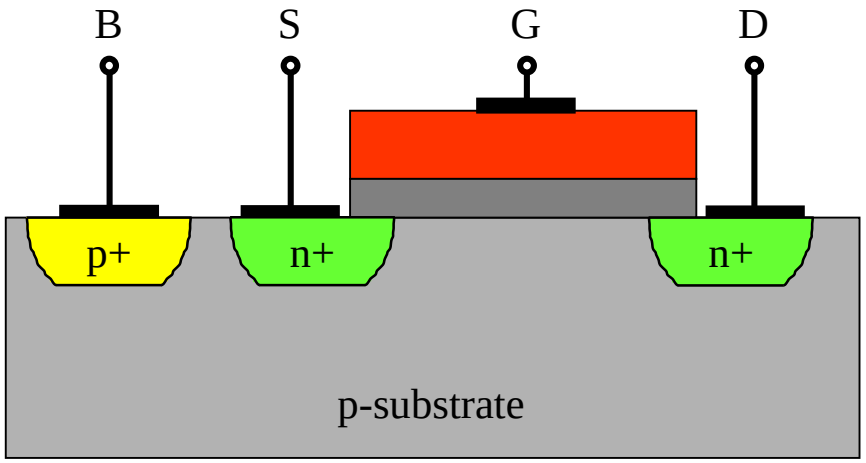
$$v_{\mathrm{GS}} < V_{\mathrm{th}}$$

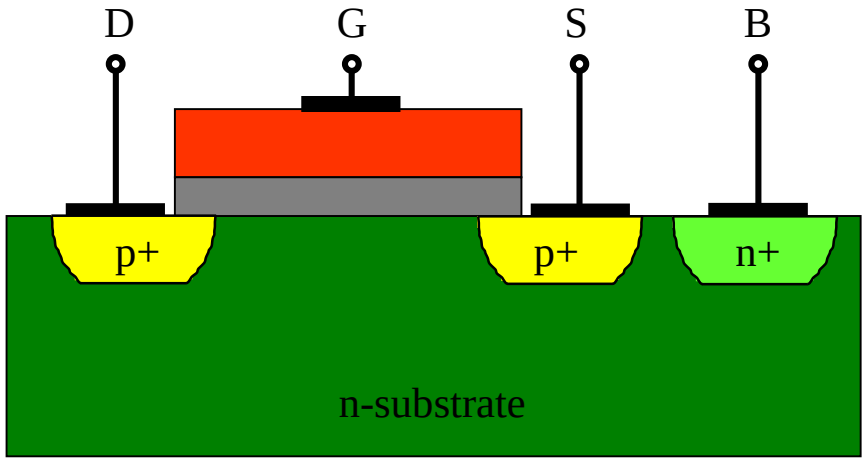


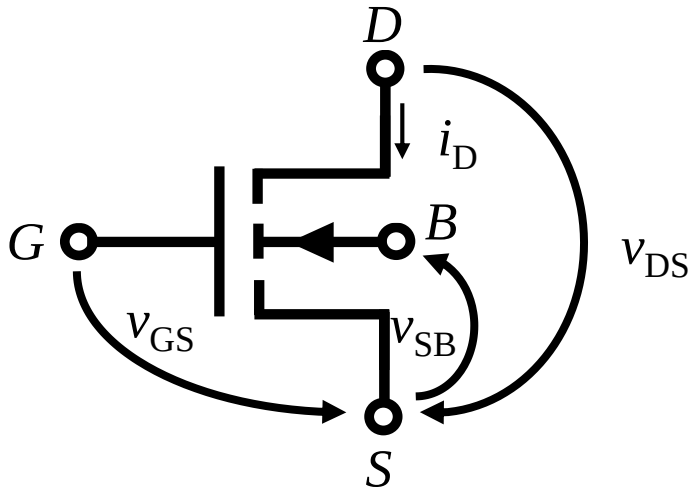
Inversion layer is established, current may flow.

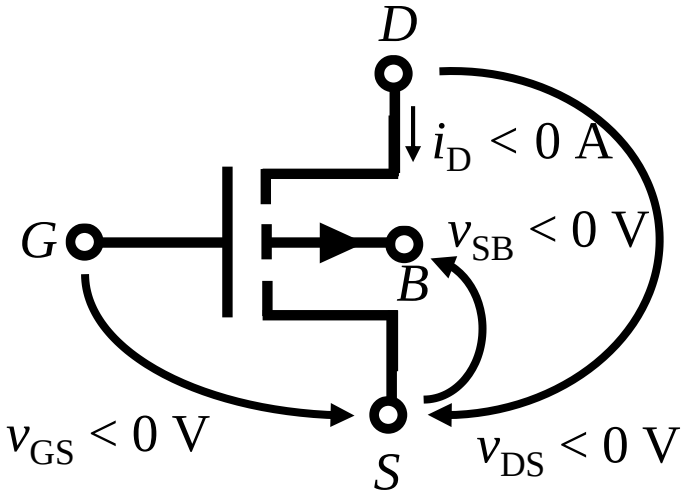
As for nMOS transistor, a linear and saturation region are to be distinguished.

nMOS versus pMOS Transistor









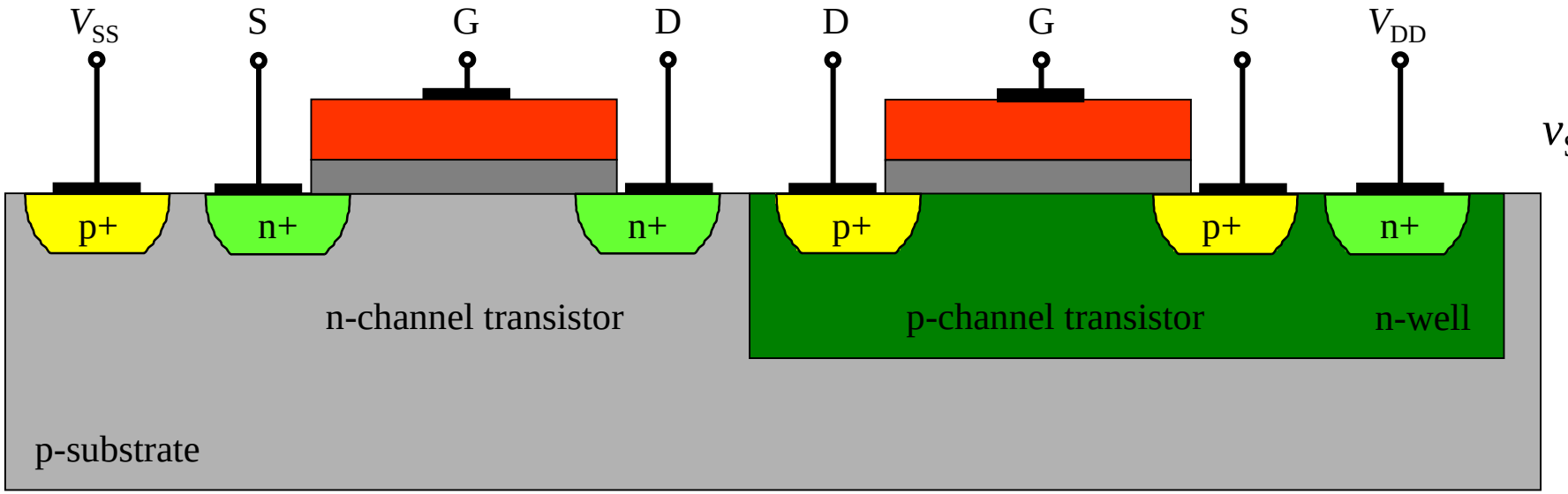
linear region

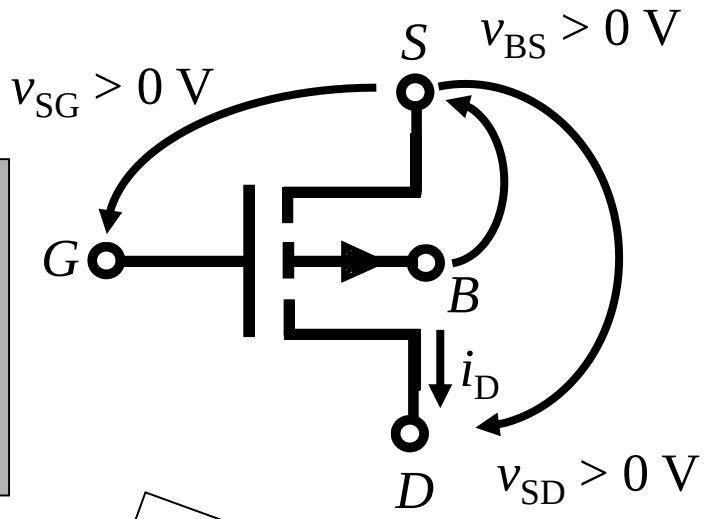
$$i_D = \beta(|v_{GS}| - |V_{th}|)|v_{DS}| - \frac{v_{DS}^2}{2}$$

saturation region

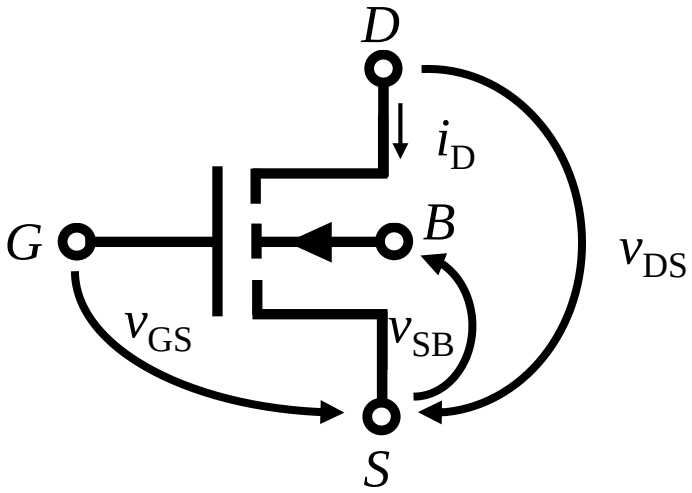
$$i_D = \frac{\beta}{2}(|v_{GS}| - |V_{th}|)^2$$

CMOS Technology – Single-Well Process





Think positive!

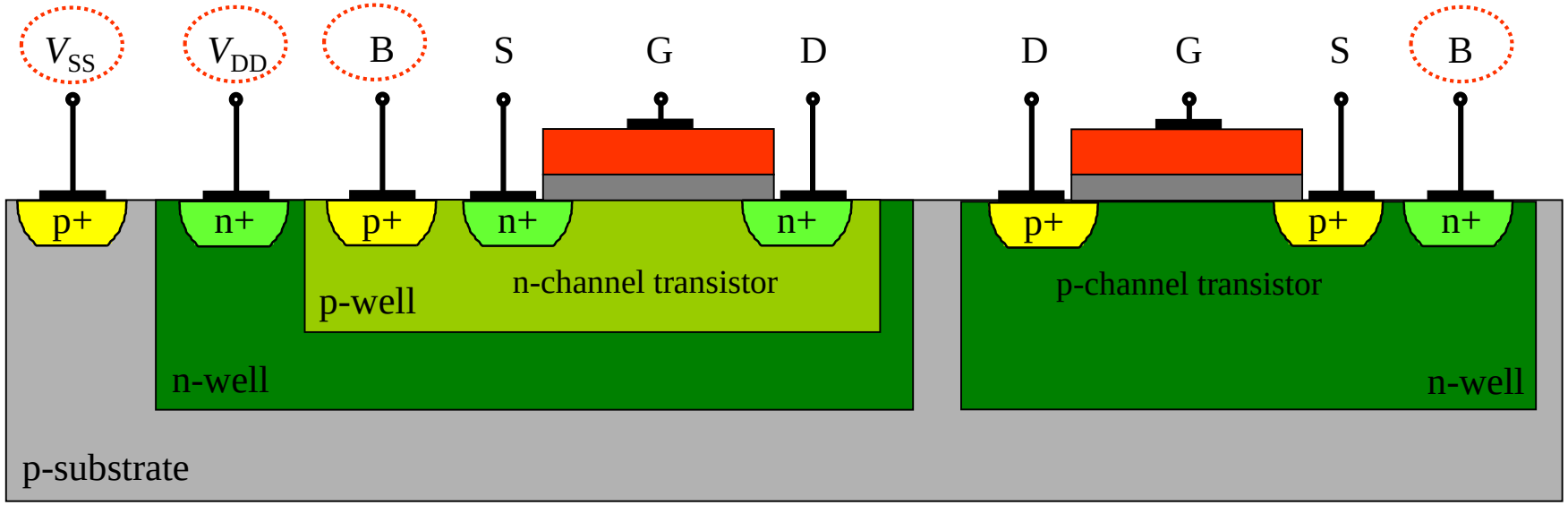


$$i_D = \frac{\beta}{2}(v_{SG} - |V_{th}|)^2$$

Use the same device equations with absolute voltage values, e.g., $|v_{GS}|$!

CMOS Technology – Twin-Well Process

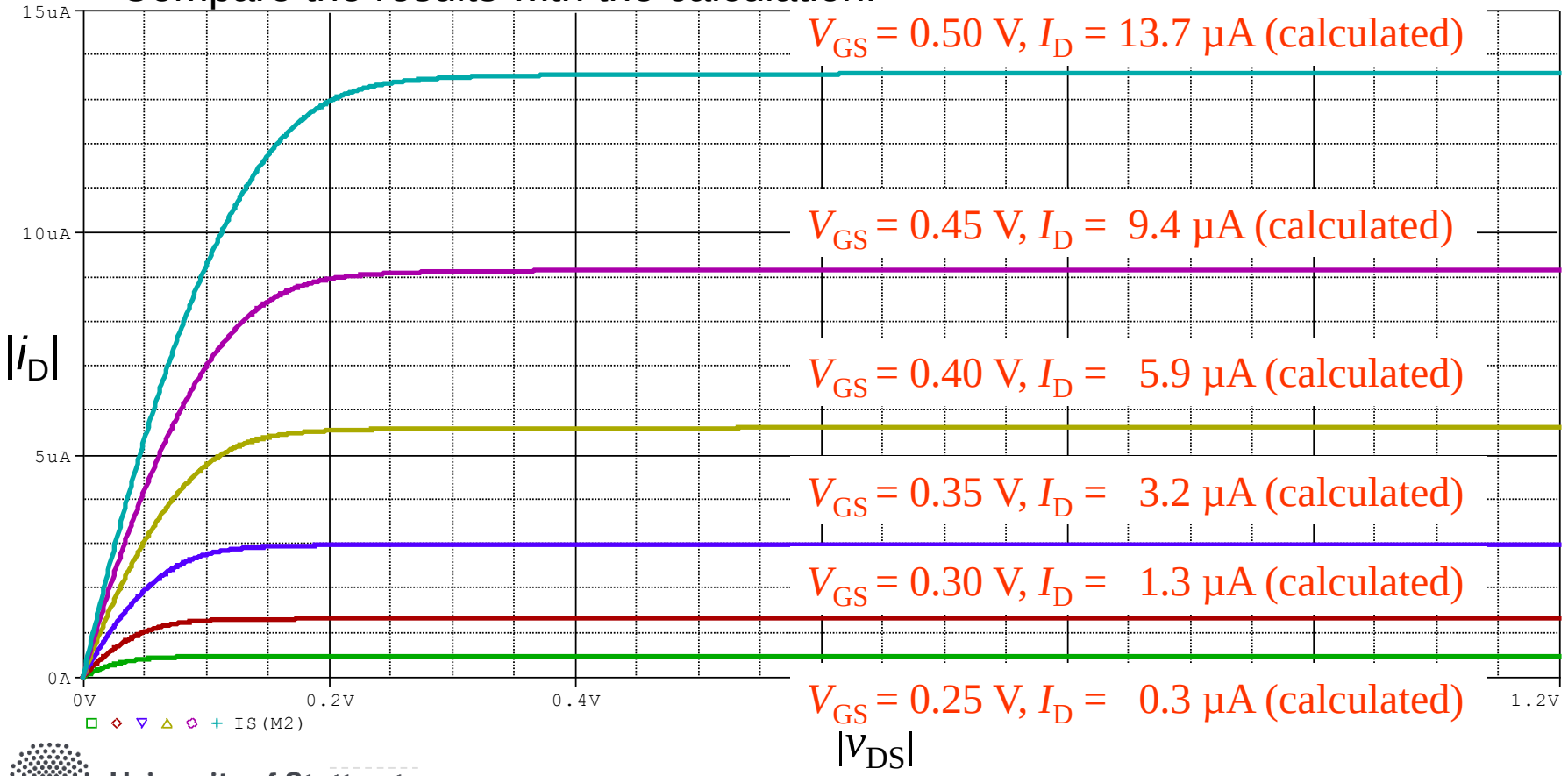
Each transistor exhibits a separate bulk contact!

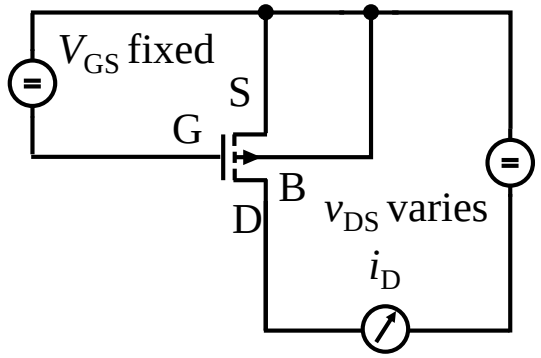


Sometimes also called a “Triple-Well Process” since the p-substrate is often realized as an epitaxial layer and thus considered as a “global well”.

Case Study – $i_D(v_{DS})$ - Transfer Characteristic (CMOS 0.12 µm)

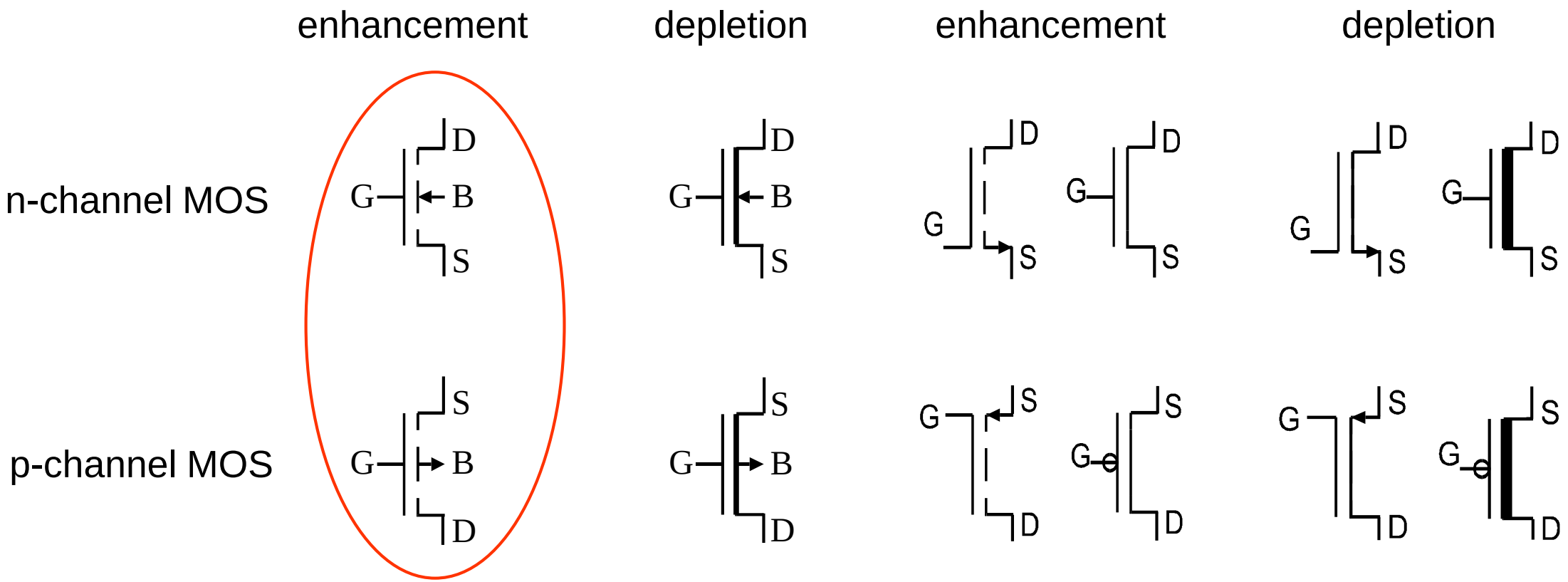
- 0.12 µm CMOS technology, $V_{thp} = 0.21\ \text{V}, \beta_{0p} = 65\ \mu\text{A/V}^2$
- Simulate i_D versus v_{DS} of a $\frac{W}{L} = \frac{100}{20}$ device for different V_{GS}!
- Compare the results with the calculation!





$$i_D = \frac{\beta_{0p}}{2}\frac{W}{L}\left(|v_{GS}| - |V_{thp}|\right)^2$$

MOS-Transistor – Symbols

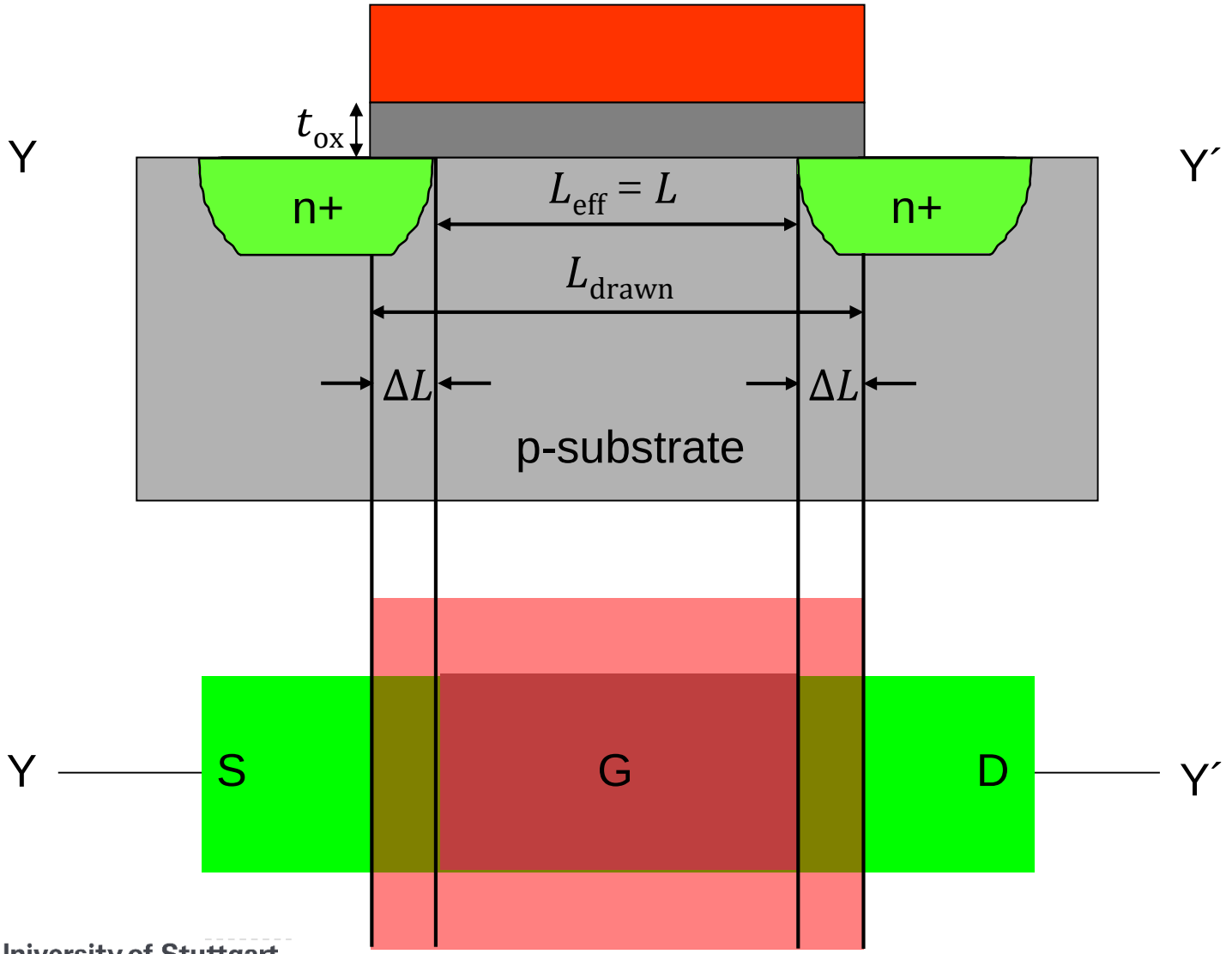


Used in our courses!

Overview

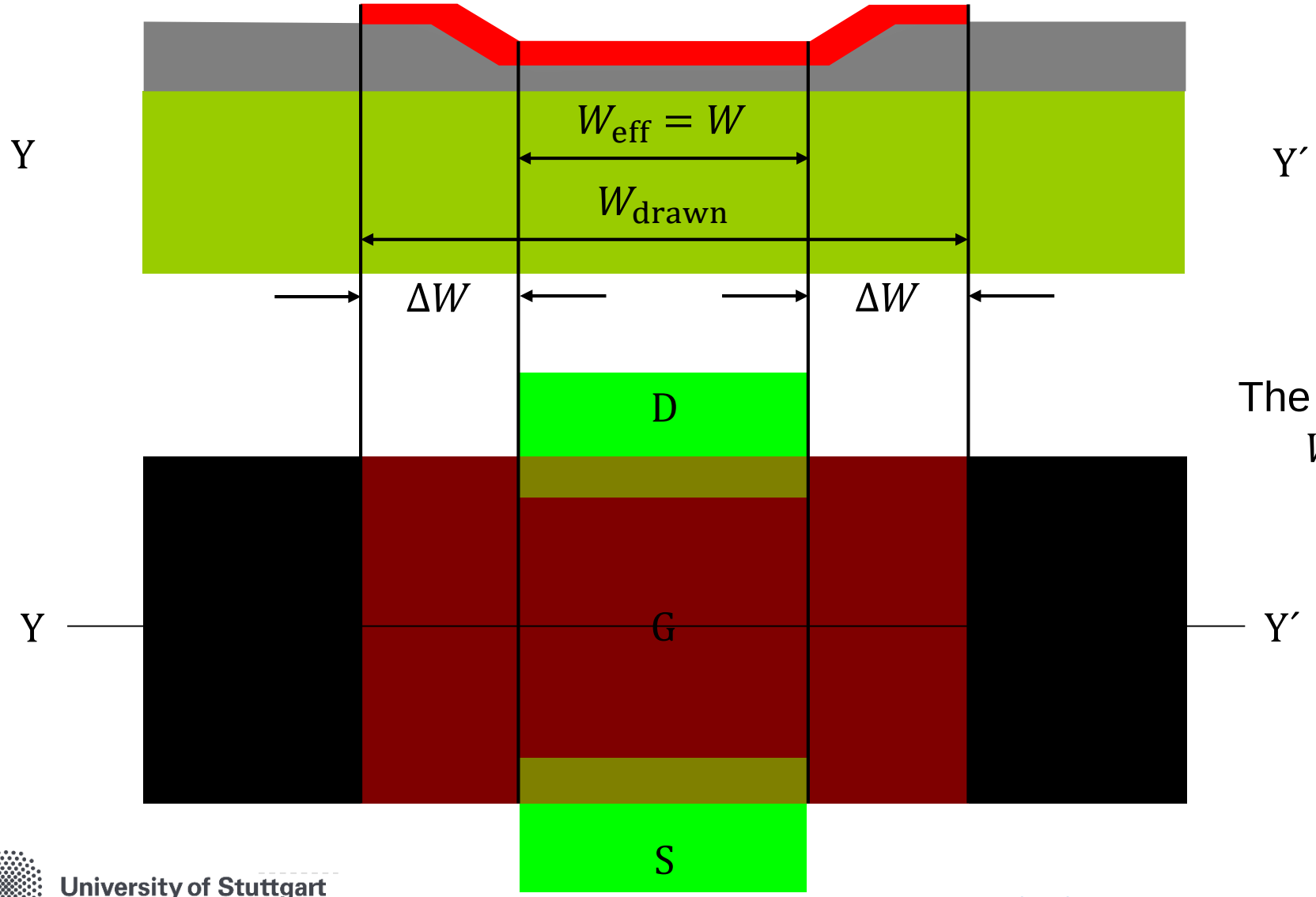
- **MOS Transistor**
  - $i(v)$ – Characteristics
  - **Second-order Effects**
    - **Effective Channel Length & Width**
    - Channel-length Modulation
    - Body Effect
    - Subthreshold Current
    - Latch-up
    - Capacitances
  - Short-channel Effects
- Passive Devices & Wiring

Effective Channel Length



The design software determines L_{drawn} so that $L_{eff} = L$.

Effective Channel Width



The design software determines W_{drawn} so that $W_{\mathrm{eff}} = W$.

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    - **Channel-length Modulation**
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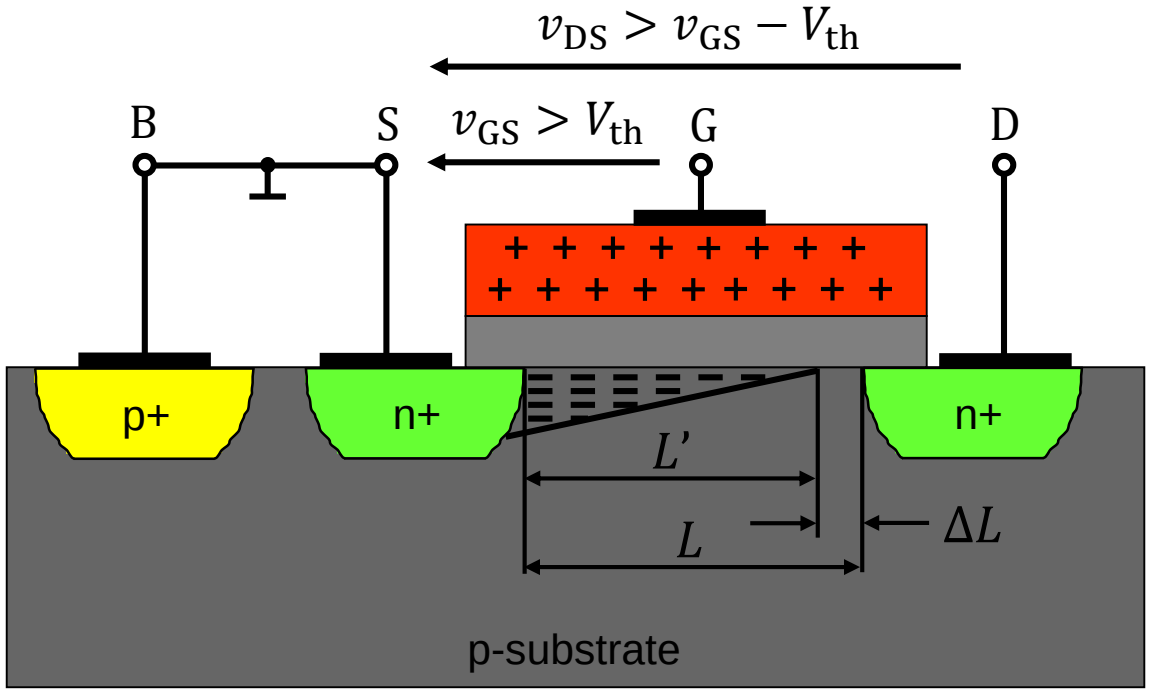
Channel Length Modulation

Saturation Region

$v_{GS} > V_{th}$

$v_{DS} > v_{GS} - V_{th}$

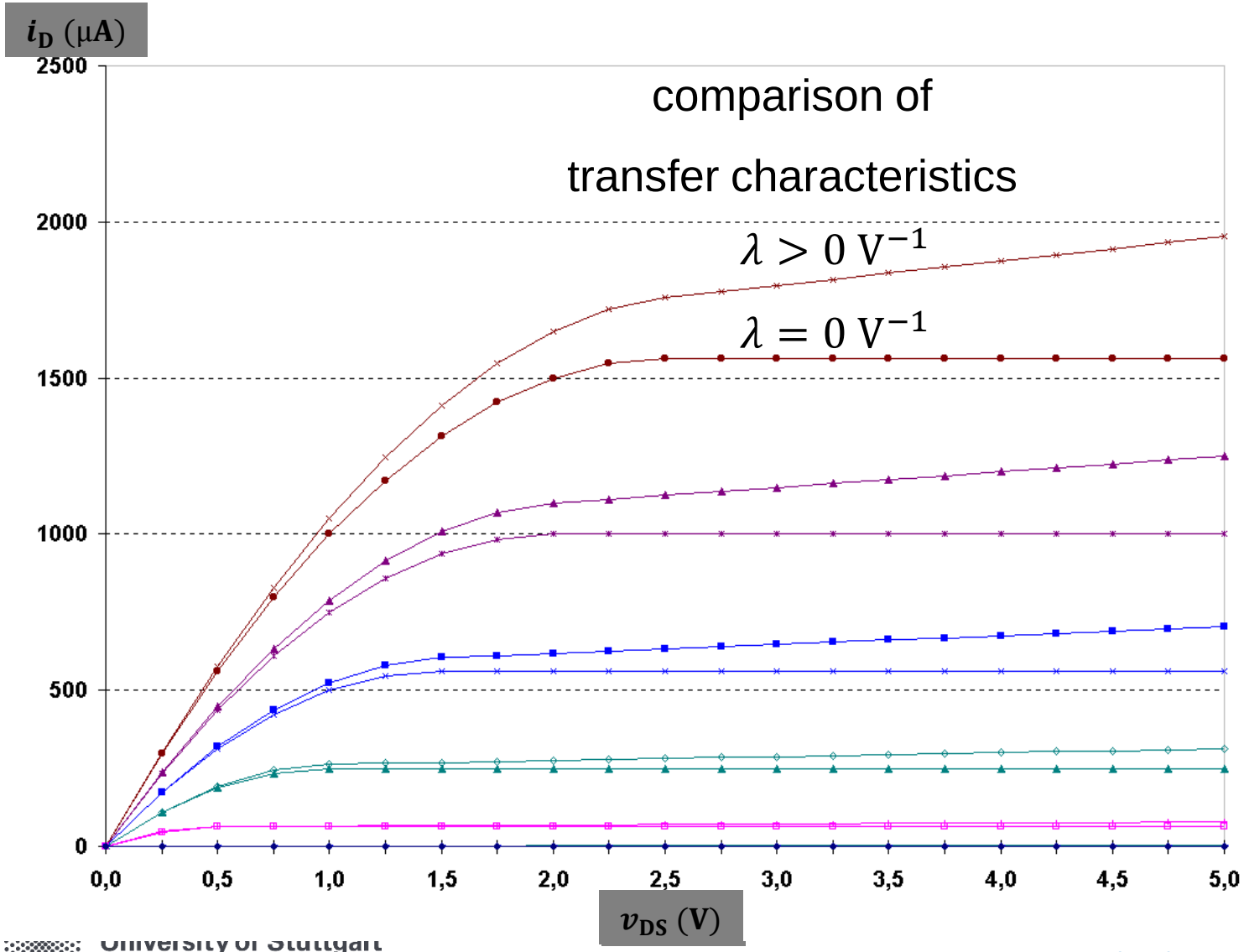
$$i_D = \frac{\mu C'_{ox} W}{2 L'} (v_{GS} - V_{th})^2$$



Replacing L' by $L - \Delta L$, performing a Taylor approximation, and assuming that $\frac{\Delta L}{L} = \lambda v_{DS}$ yields: $i_D = \frac{\beta}{2}(v_{GS} - V_{th})^2(1+\lambda v_{DS})$

i_D slightly increases with v_{DS} in saturation!

$i_D(v_{DS})$ - Transfer Characteristic



saturation: $v_{DS} > v_{GS} - V_{th}$

$$i_D = \frac{\beta}{2}(v_{GS} - V_{th})^2(1+\lambda v_{DS})$$

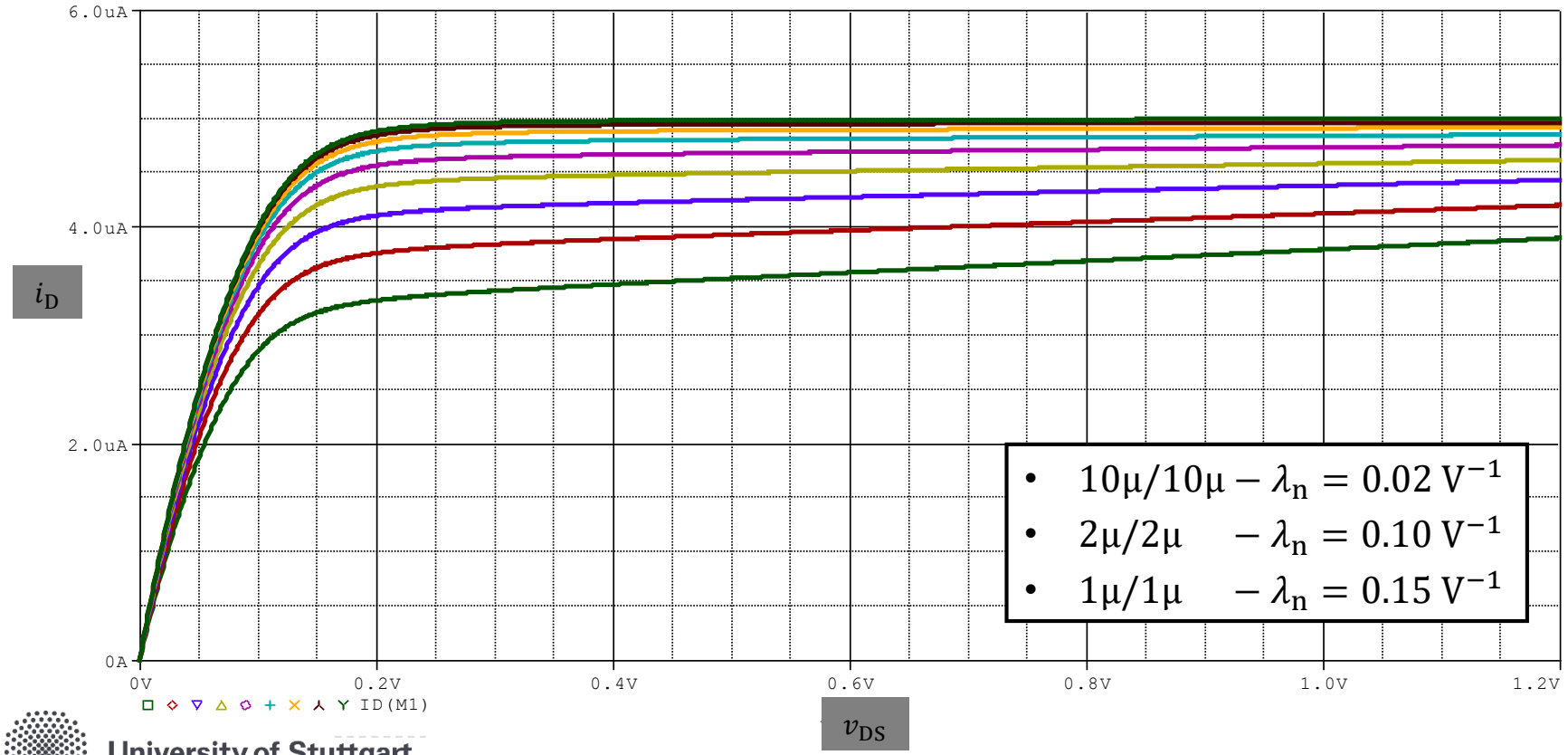
linear: $v_{DS} < v_{GS} - V_{th}$

$$i_D = \beta\left(v_{GS} - V_{th} - \frac{1}{2}v_{DS}\right)v_{DS}(1+\lambda v_{DS})$$

$i_\mathrm{D}(v_\mathrm{DS})$ - nMOS Transfer Characteristic – Determination λ

0.12µm CMOS (nMOS long-channel): $V_\mathrm{th} = 0.19\ \mathrm{V}$, $\beta_0 = 370\ \mu\mathrm{A/V}^2$

Simulation of a $W/L = 1\mu\mathrm{m}/1\mu\mathrm{m} \ldots 100\mu\mathrm{m}/100\mu\mathrm{m}$ device for $V_\mathrm{GS} = 0.35\ \mathrm{V}$



$$I_\mathrm{D1} = \frac{\beta}{2}(V_\mathrm{GS} - V_\mathrm{th})^2(1 + \lambda V_\mathrm{DS1})$$

$$I_\mathrm{D2} = \frac{\beta}{2}(V_\mathrm{GS} - V_\mathrm{th})^2(1 + \lambda V_\mathrm{DS2})$$

$$\lambda = \frac{I_\mathrm{D2} - I_\mathrm{D1}}{I_\mathrm{D1}V_\mathrm{DS2} - I_\mathrm{D2}V_\mathrm{DS1}}$$

The longer the device, the smaller λ!

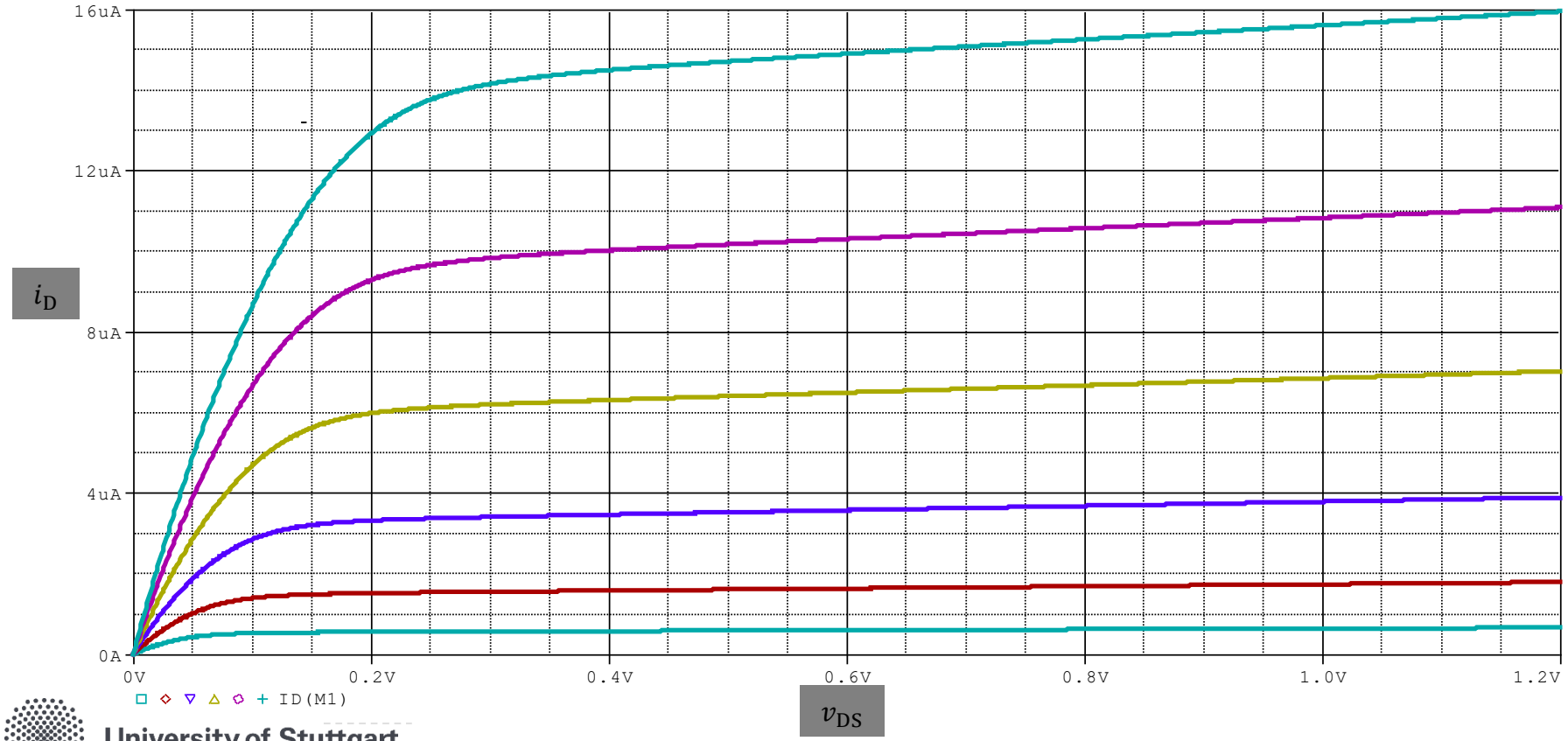
$$\lambda \sim \frac{\Delta L}{L}$$

$i_\mathrm{D}(v_\mathrm{DS})$ - nMOS Transfer Characteristic

0.12µm CMOS (nMOS long-channel): $V_\mathrm{th} = 0.19\ \mathrm{V}$, $\beta_0 = 370\ \mu\mathrm{A/V}^2$

Simulation of a $\frac{W}{L} = 1\mu\mathrm{m}/1\mu\mathrm{m}$ device for $v_\mathrm{GS} = 0.25 \ldots 0.5\ \mathrm{V}$

$$i_\mathrm{D} = \frac{\beta}{2}(v_\mathrm{GS} - V_\mathrm{th})^2(1 + \lambda v_\mathrm{DS})$$

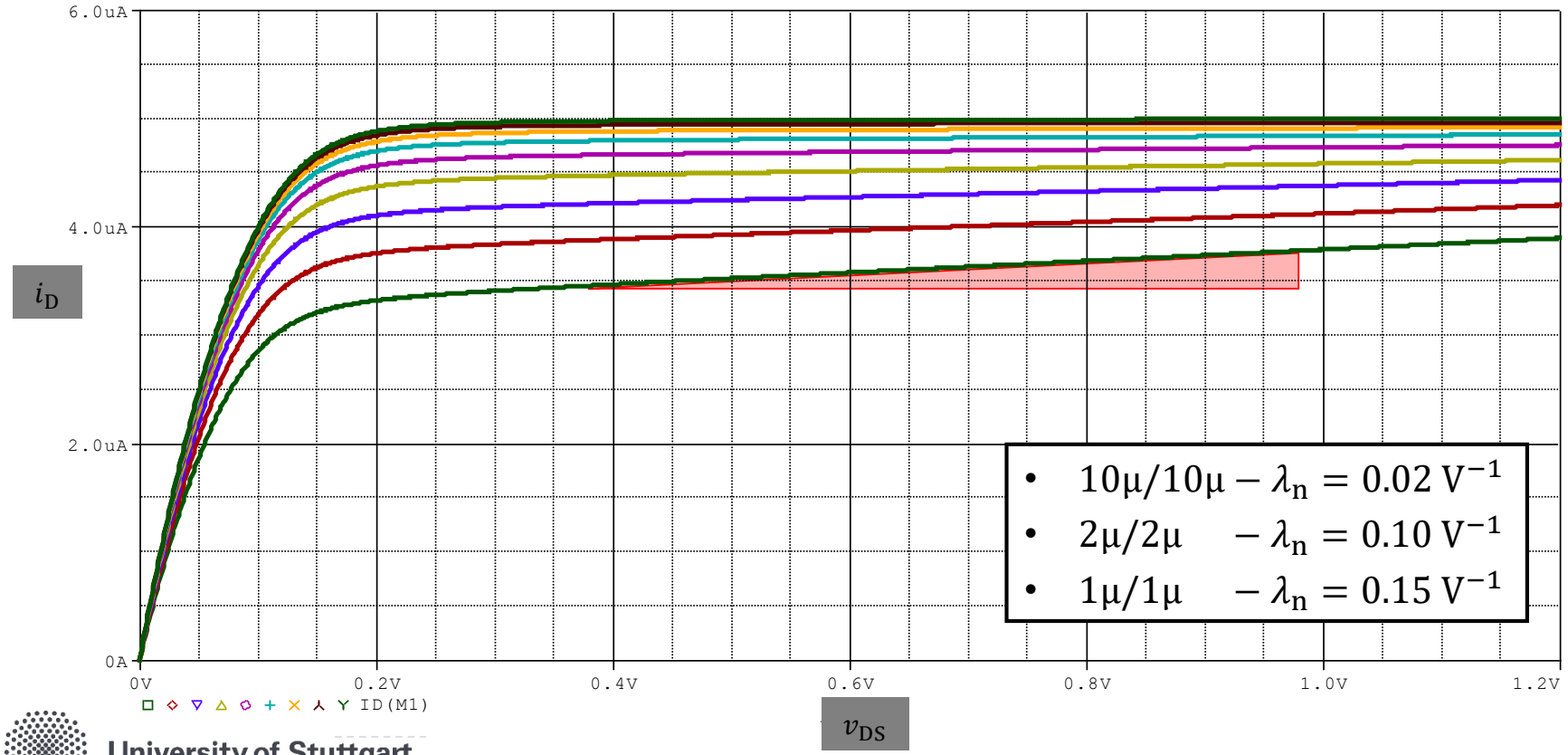


Slope varies for different i_D $\rightarrow\ \lambda(v_\mathrm{GS})$

$i_\mathrm{D}(v_\mathrm{DS})$ - nMOS Transfer Characteristic – Slope

0.12µm CMOS (nMOS long-channel): $V_\mathrm{th} = 0.19\ \mathrm{V}$, $\beta_0 = 370\ \mu\mathrm{A/V^2}$

Simulation of a $W/L = 1\mu\mathrm{m}/1\mu\mathrm{m} \ldots 100\mu\mathrm{m}/100\mu\mathrm{m}$ device for $V_\mathrm{GS} = 0.35\ \mathrm{V}$



$$I_\mathrm{D1} = \frac{\beta}{2}(V_\mathrm{GS} - V_\mathrm{th})^2(1 + \lambda V_\mathrm{DS1})$$

$$I_\mathrm{D2} = \frac{\beta}{2}(V_\mathrm{GS} - V_\mathrm{th})^2(1 + \lambda V_\mathrm{DS2})$$

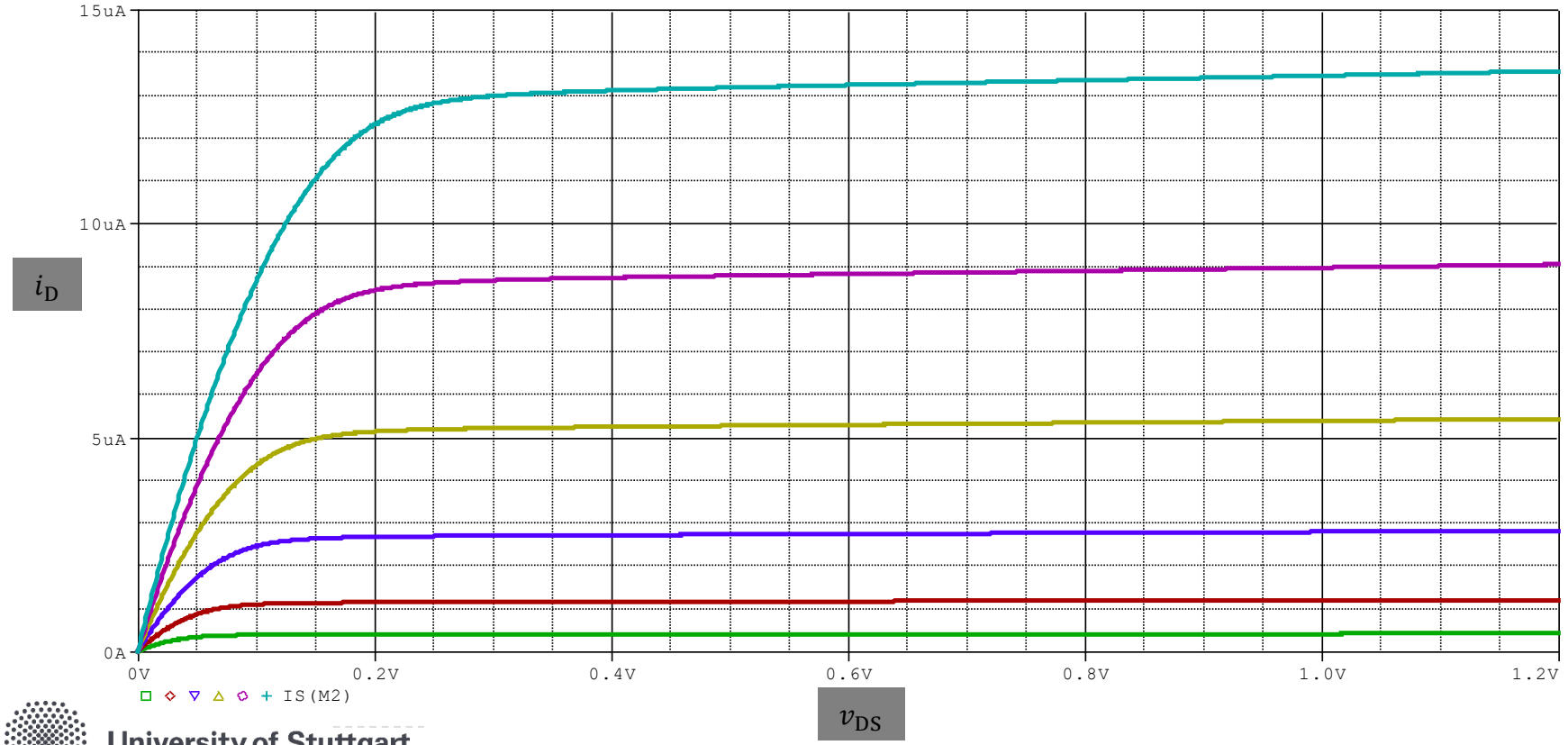
The slope in saturation results in

$$\frac{I_\mathrm{D2} - I_\mathrm{D1}}{V_\mathrm{DS2} - V_\mathrm{DS1}} = \lambda \frac{\beta}{2}(V_\mathrm{GS} - V_\mathrm{th})^2$$

$i_\mathrm{D}(v_\mathrm{DS})$ - pMOS Transfer Characteristic

0.12µm CMOS (nMOS long-channel): $V_\mathrm{th} = 0.21\ \mathrm{V}$, $\beta_0 = 65\ \mu\mathrm{A/V^2}$

Simulation of a $\frac{W}{L} = 5\mu\mathrm{m}/1\mu\mathrm{m}$ device for $v_\mathrm{GS} = 0.25 \ldots 0.5\ \mathrm{V}$



The longer the device, the smaller λ!

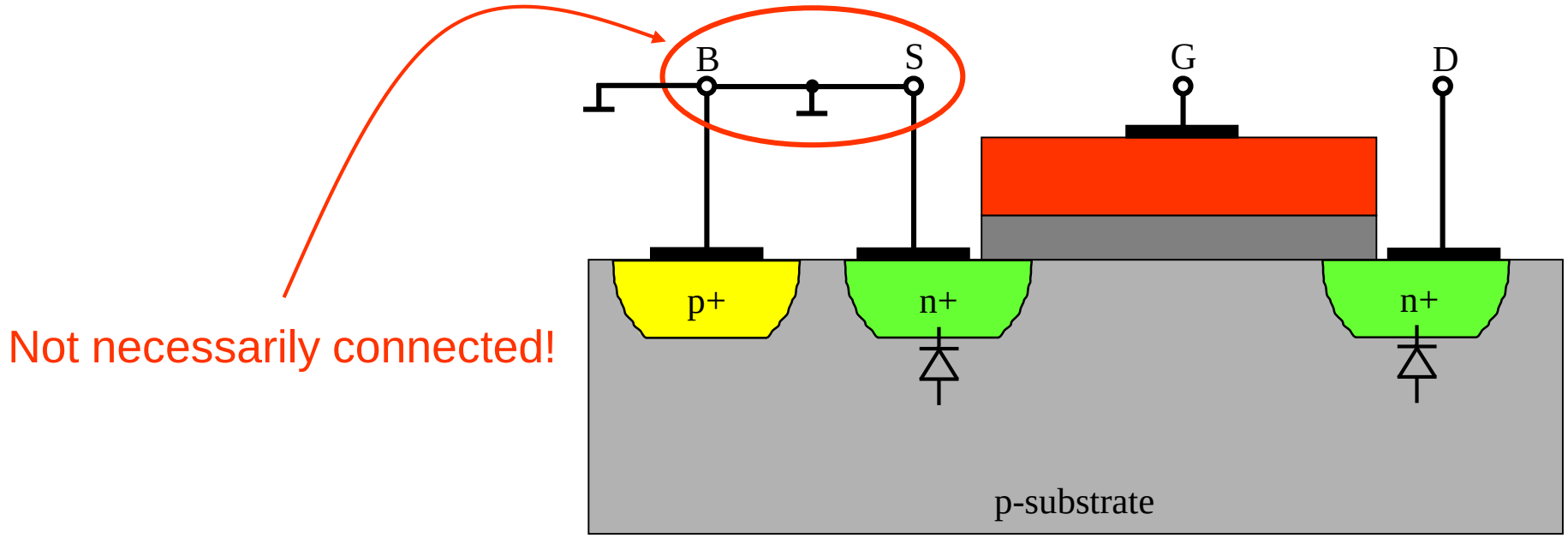
$$\lambda \sim \frac{\Delta L}{L}$$

- $10\mu/10\mu - \lambda_\mathrm{p} = 0.005\ \mathrm{V^{-1}}$
- $10\mu/2\mu - \lambda_\mathrm{p} = 0.020\ \mathrm{V^{-1}}$
- $1\mu/1\mu - \lambda_\mathrm{p} = 0.040\ \mathrm{V^{-1}}$

Overview

- **MOS Transistor**
  - $i(v)$ – Characteristics
  - **Second-order Effects**
    - Effective Channel Length & Width
    - Channel-length Modulation
    - **Body Effect**
    - Subthreshold Current
    - Latch-up
    - Capacitances
  - Short-channel Effects
- Passive Devices & Wiring

Body Effect

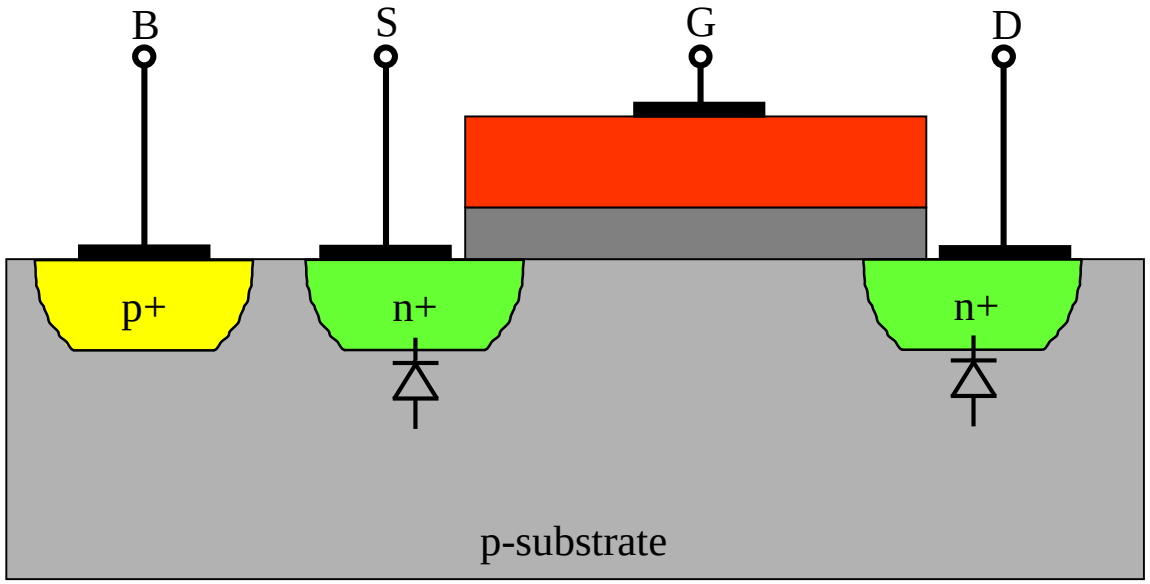


The bulk (B) of an nMOS (pMOS) transistor is usually connected to the lowest (highest) potential in order to keep the SB and DB diodes reverse biased!

nMOS: $v_B \leq v_S \leq v_D$

pMOS: $v_B \geq v_S \geq v_D$

Body Effect – Equation

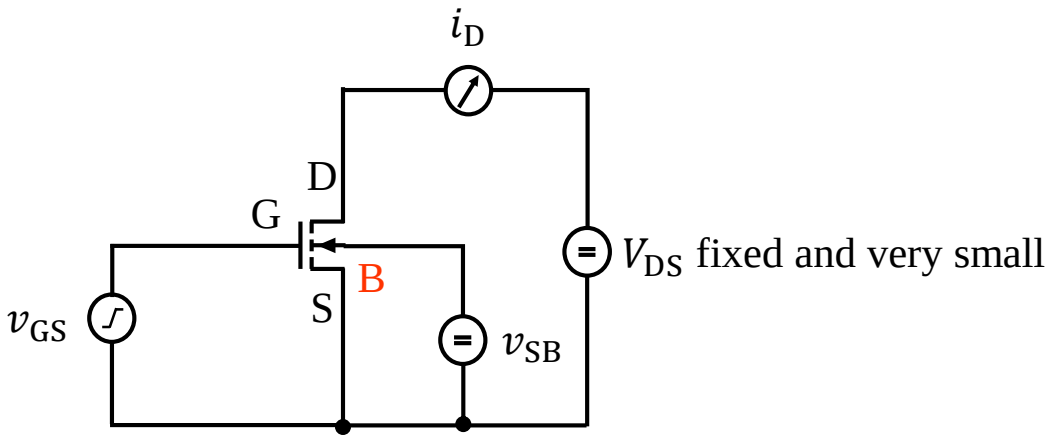


The threshold voltage always increases if $v_{\mathrm{SB}} \neq 0\ \mathrm{V}$!

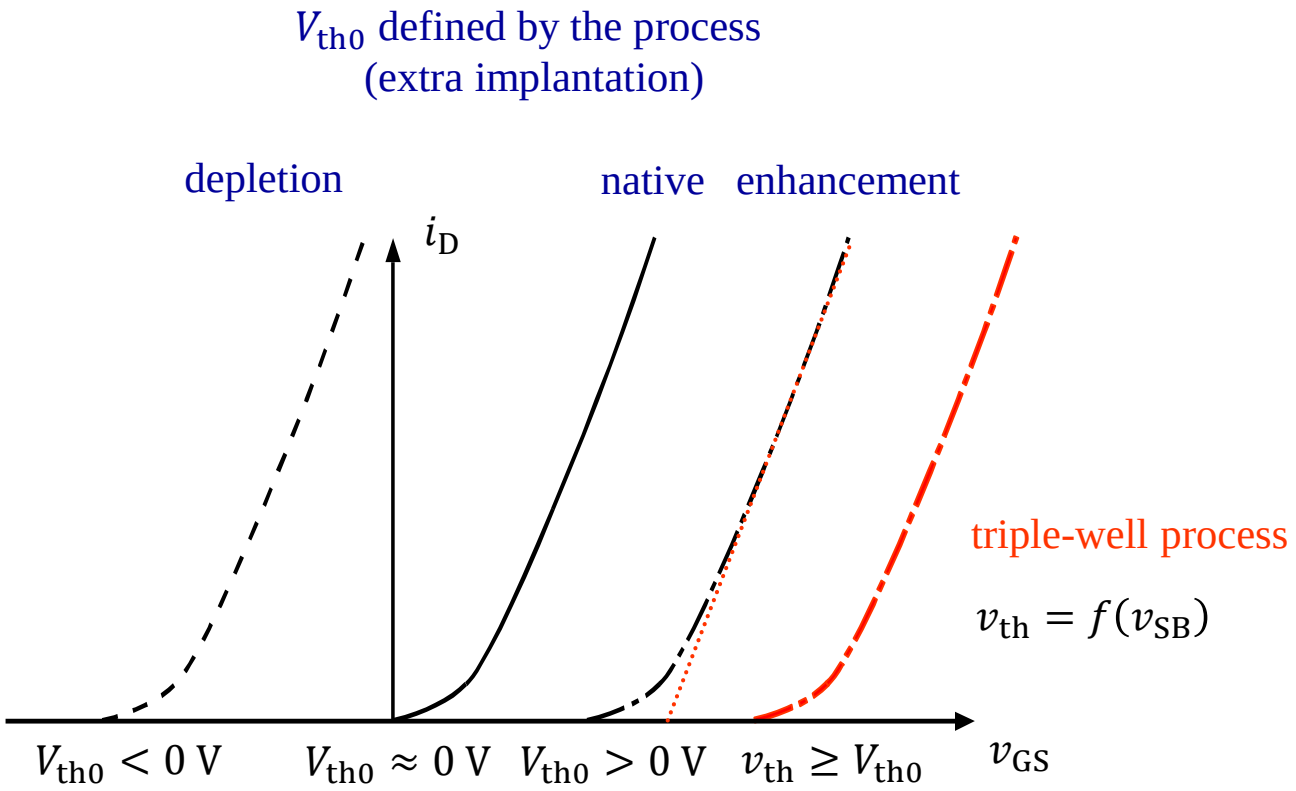
$$v_{\mathrm{th}} = V_{\mathrm{th0}} + \gamma\left(\sqrt{2\Phi_{\mathrm{F}} + v_{\mathrm{SB}}} - \sqrt{2\Phi_{\mathrm{F}}}\right)$$

γ – body-effect coefficient $\quad \gamma = \dfrac{\sqrt{2\varepsilon_{\mathrm{Si}} q N_{\mathrm{sub}}}}{C'_{\mathrm{ox}}} \qquad \Phi_{\mathrm{F}} = \dfrac{kT}{q}\ln\left(\dfrac{N_{\mathrm{sub}}}{n_{\mathrm{i}}}\right)$

$i_D(v_{GS})$ - Transfer Characteristic



$$v_{th} \approx V_{th0} + \gamma\sqrt{v_{SB}}$$ Body effect!



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Subthreshold Current

assumption: $v_{GS} < V_{th}$, $v_{DS} > 0$ $\quad i_D = 0\,A$?

$$i_D = I_0 e^{\frac{v_{GS}-V_{th}}{\zeta V_T}}\left(1 - e^{\frac{-v_{DS}}{V_T}}\right) \quad \text{with} \quad V_T = \frac{kT}{q}, \zeta > 1, I_0 = \mu_n C'_D \frac{W}{L} V_T^2$$

for $v_{DS} > 200\,mV$

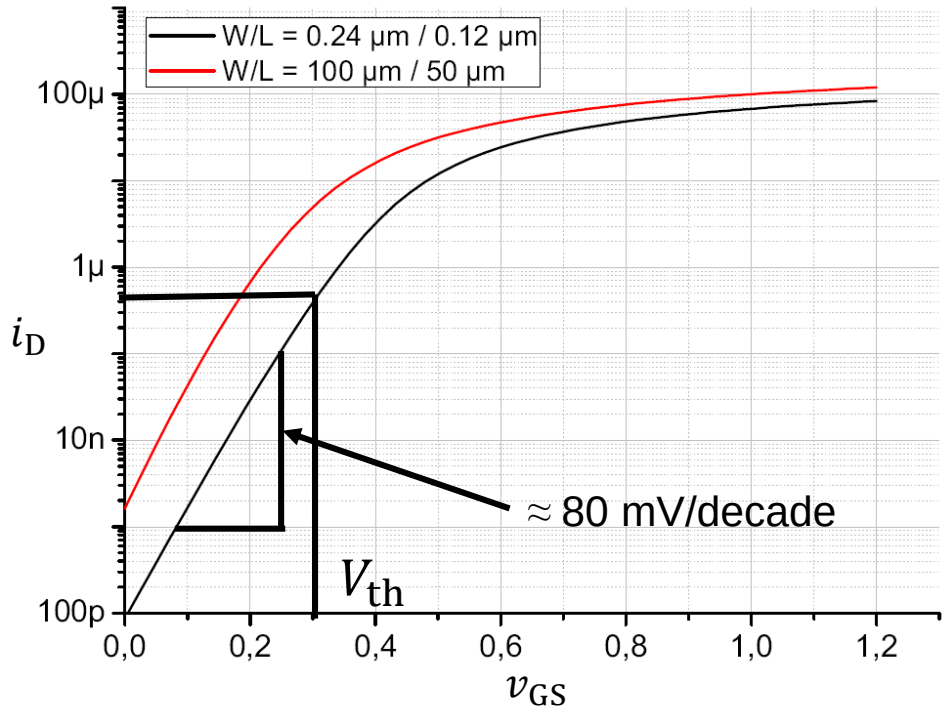
$$i_D \approx I_0 e^{\frac{v_{GS}-V_{th}}{\zeta V_T}}$$

There is current flowing for $v_{GS} < V_{th}$!

However, it is very small!

→ subthreshold circuit design

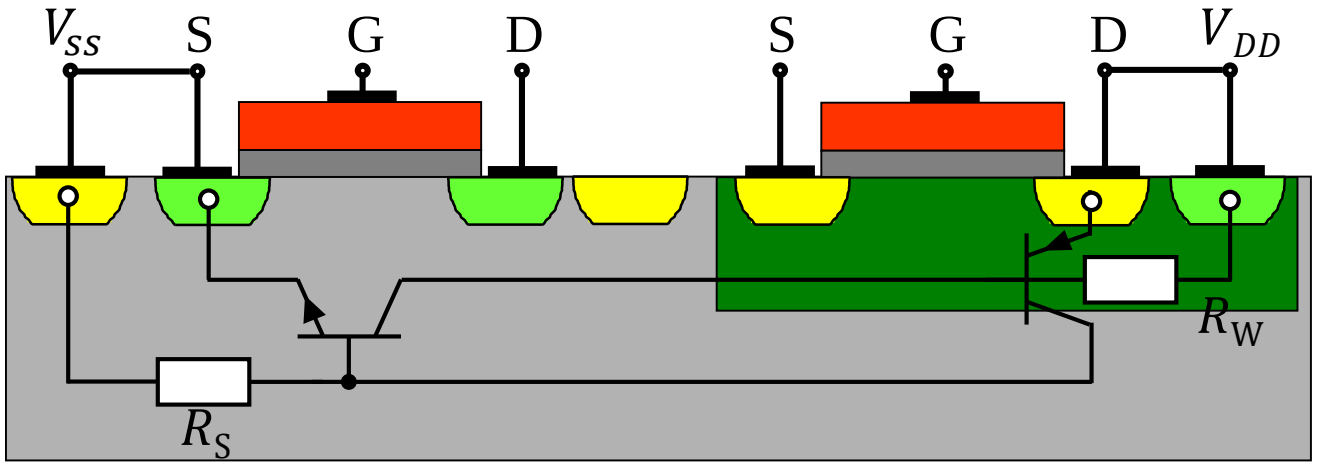
(no topic of this lecture)

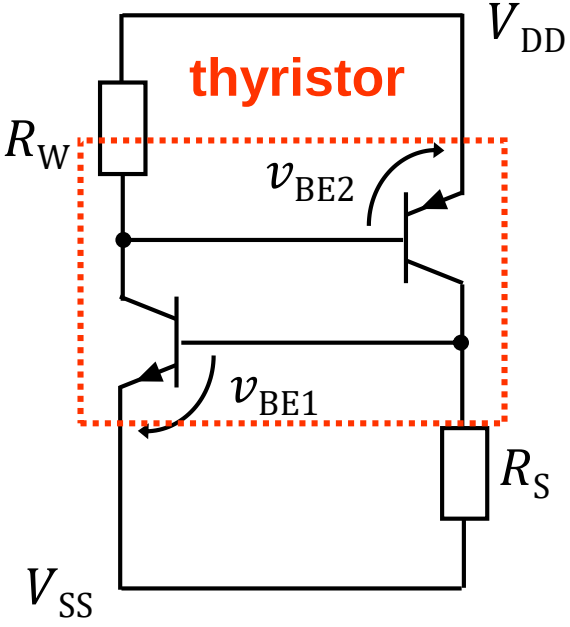


Overview

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    - Capacitances
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Latch-up





formation:

voltage drop on R_W
→ v_{BE2} turns on
→ voltage drop on R_S
→ v_{BE1} turns on
→ voltage drop on R_W increases…

reason:

peaks in V_{DD} and/or V_{SS}

requirement: $\beta_1\beta_2 > 1$

protective measures:

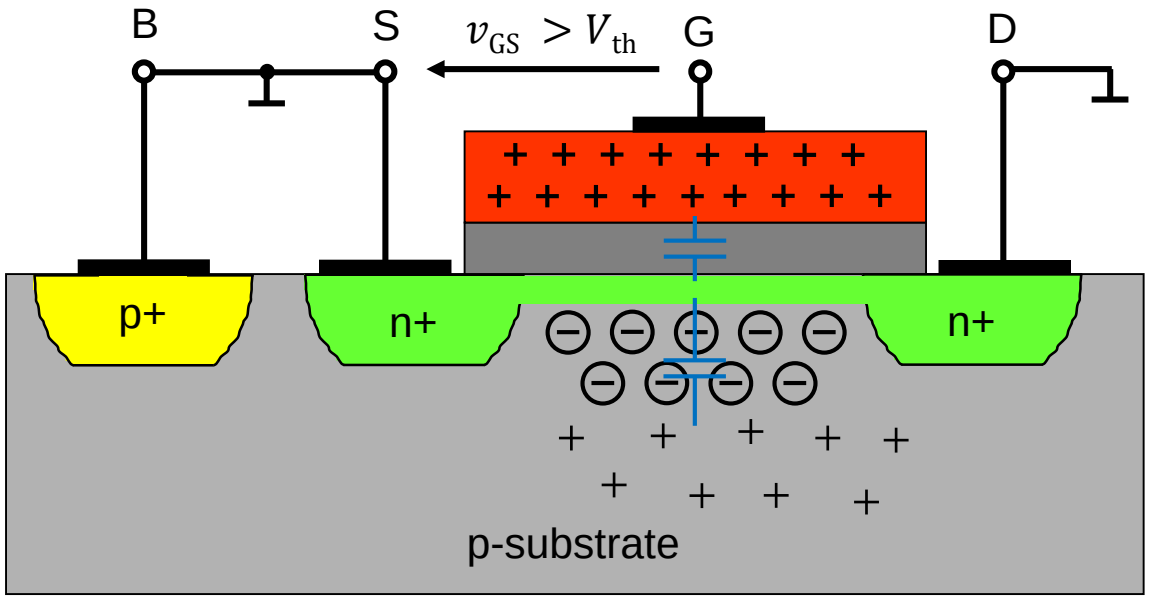
process: * highly doped buried layer → $R_s\downarrow$, $R_w\downarrow$
* oxide trenches
→ development of thyristor not feasible

circuit: * large interspace → $\beta\downarrow$
* guard rings (= many substrate/well contacts)
→ stabilization of substrate/well potential

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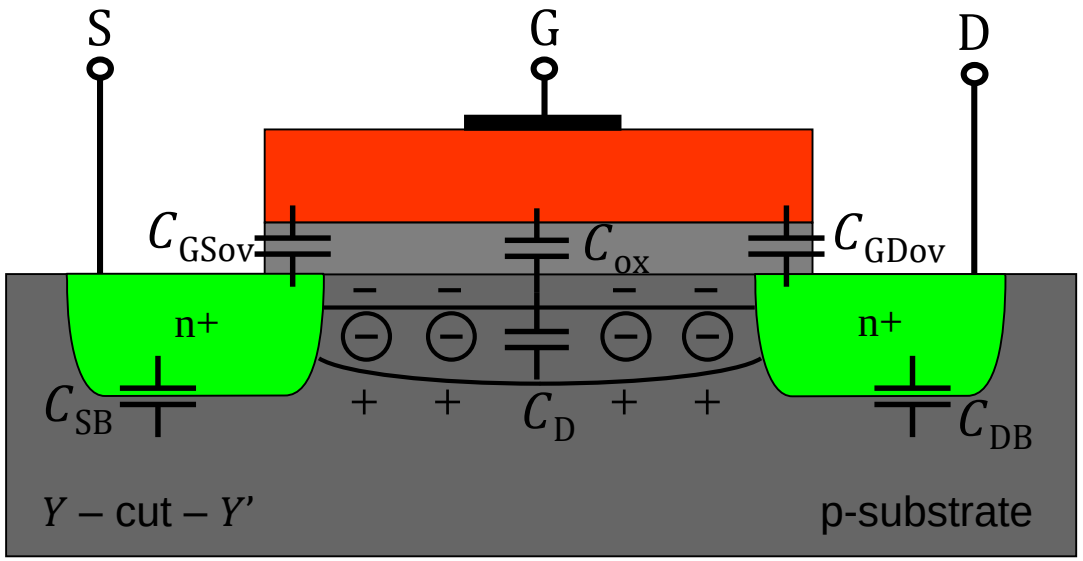
MOS Transistor – A Complicated Capacitor

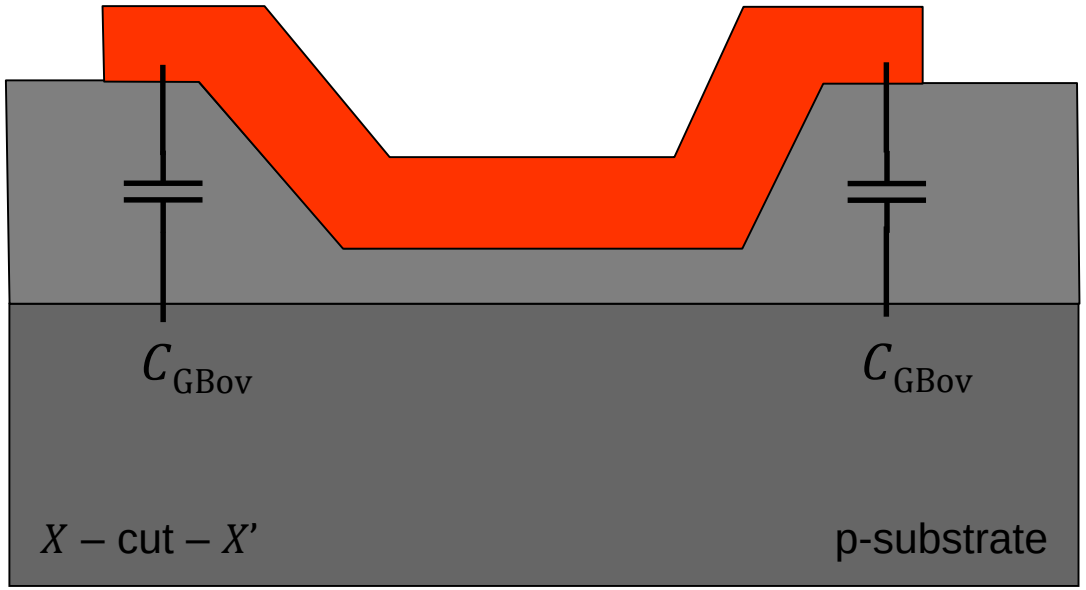


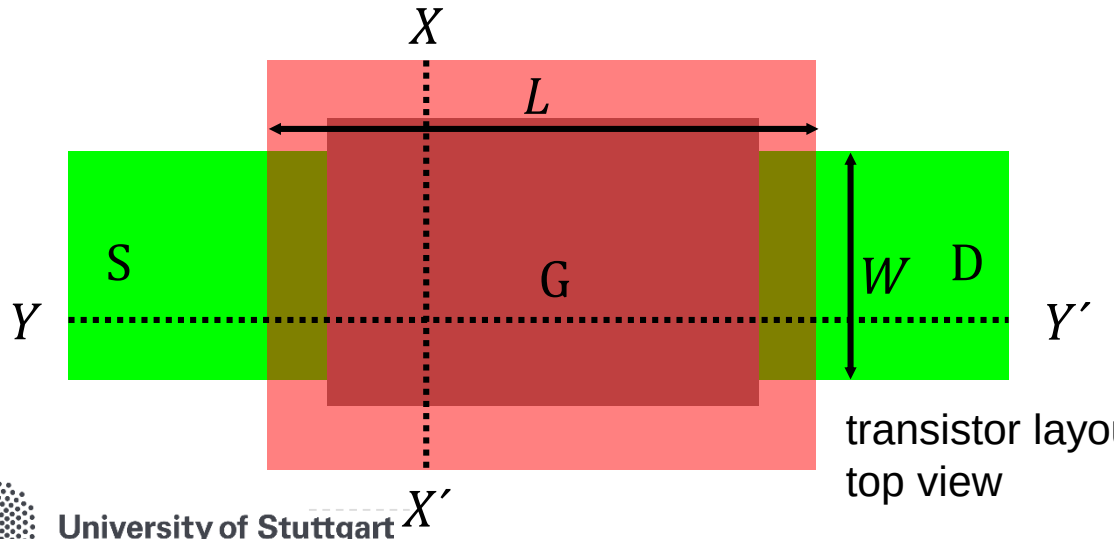
insulator → plate capacitor

depletion region (only cores,
no free charge carriers)
→ „insulator“ → depletion capacitor

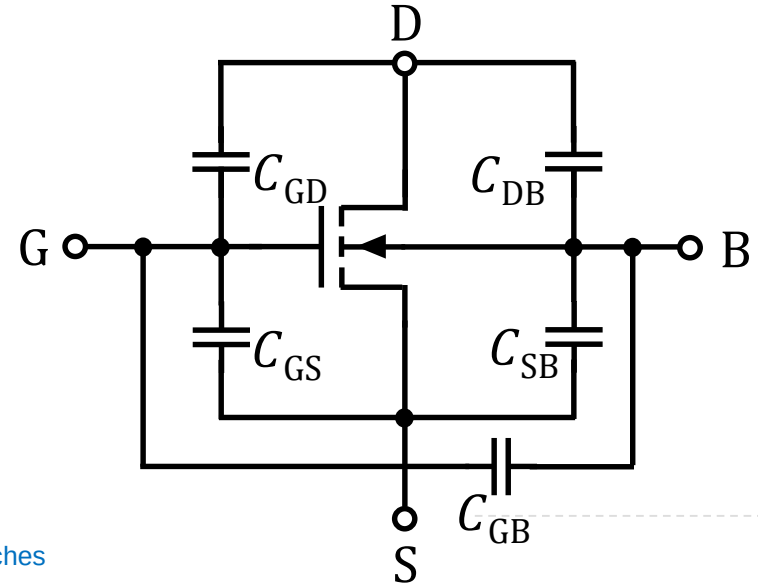
MOS Transistor – Parasitic Capacitors



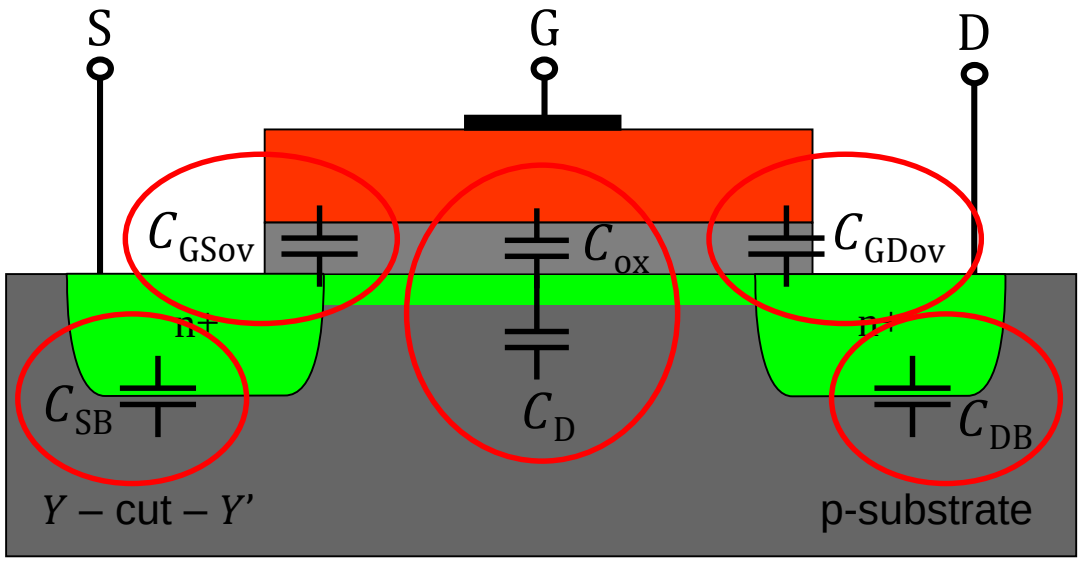


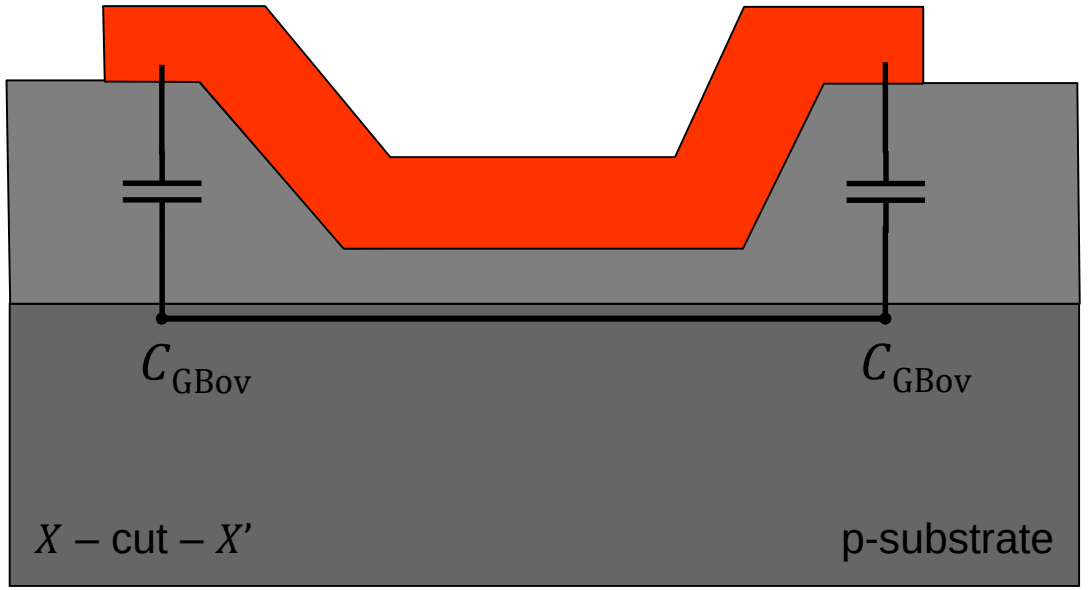


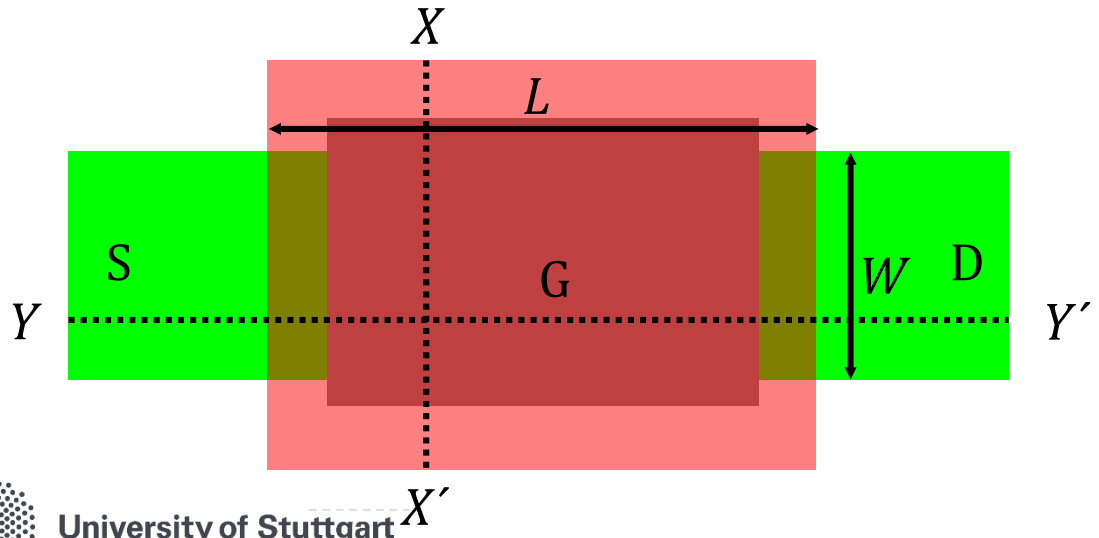
transistor layout
top view



Capacitances – Approximated Calculation







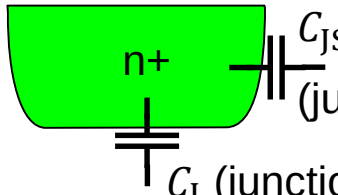
$$C_{DB} \approx C_J A_D + C_{JSW} P_D$$

$$C_{SB} \approx C_J A_S + C_{JSW} P_S$$

$$C_{GDov} \approx C_{GD0} W$$

$$C_{GSOV} \approx C_{GS0} W$$

$$C_{GBov} \approx 2 C_{GB0} L$$

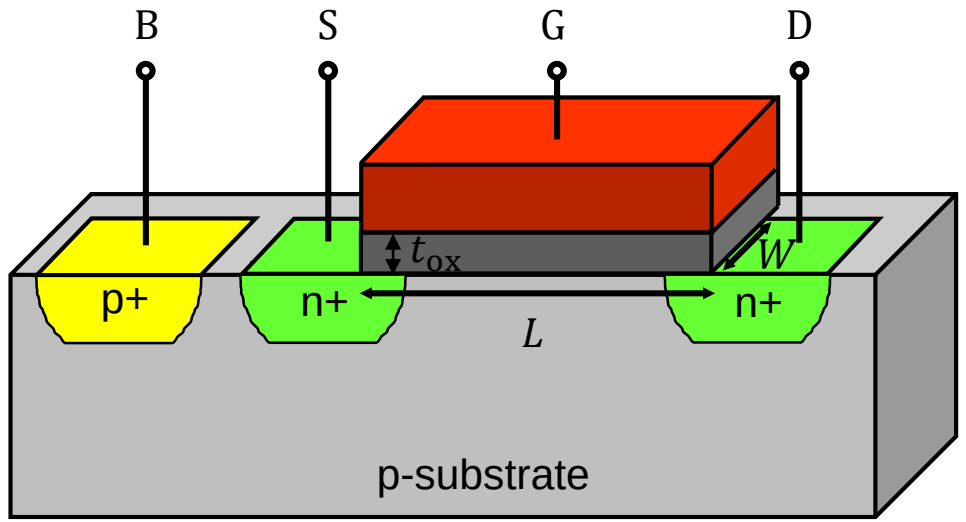


$$C_{ox} \approx C'_{ox} WL$$

$$C_D \approx \sqrt{q \epsilon_{Si} \frac{N_{SUB}}{4 \Phi_F}} WL$$

Our CMOS 0.13 µm Technology – Case Study

$$C_{ox} \approx C'_{ox} WL = \frac{\epsilon_{ox}\epsilon_0}{t_{ox}} WL$$



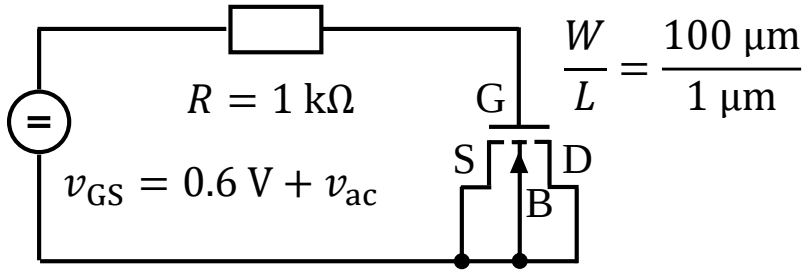
- with
  - C'_{ox} – sheet capacitance of the gate-oxide capacitor
  - W, L – width & length of the transistor
  - ϵ_{ox} – relative dielectric constant (SiO_2: 3.9)
  - ϵ_0 – permittivity of the free space ($8.854 * 10^{-12}\mathrm{Vs/Am}$ or $\mathrm{C^2/Nm^2}$)
  - t_{OX} – thickness of the insulator (our technology: $\mathrm{t_{ox}} = 2.73\ \mathrm{nm}$)

$$C'_{ox} = \frac{\epsilon_{ox}\epsilon_0}{t_{ox}} = \underbrace{0.012\,\frac{\mathrm{F}}{\mathrm{m^2}}}_{\text{typ}} = \underbrace{12\,\frac{\mathrm{fF}}{\mu\mathrm{m^2}}}_{\text{typ}}$$

We talk of µm in Microelectronics!

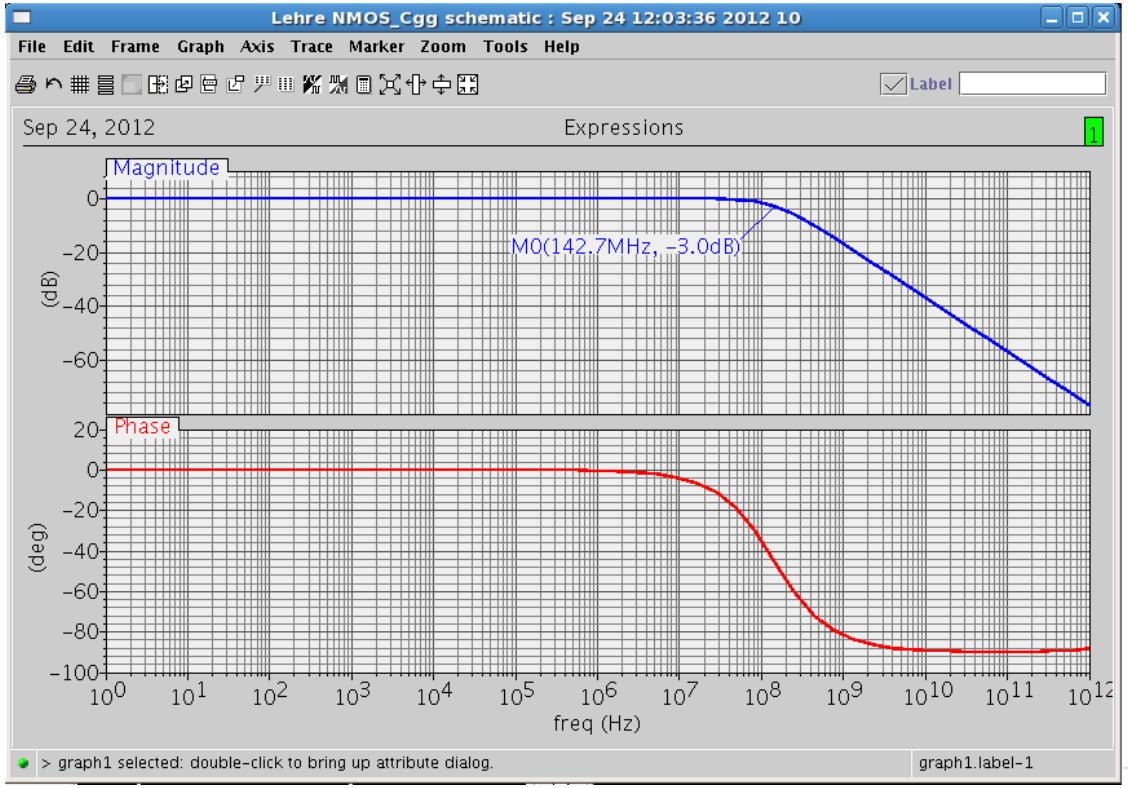
Our CMOS 0.13 µm Technology – Case Study

- How to extract the sheet capacitance C'_{ox} by simulation?
  - → Simulate the -3 dB cut-off frequency of an RC low-pass filter (C implemented by C_{ox} of the transistor)



- $\omega = \frac{1}{RC}$
- $f_c = 142.7\ \mathrm{MHz}$ (simulated)
- $\omega_c = \frac{1}{RC} = 2\pi f_c = 896\ \mathrm{Mrad/s}$
- $C = 1.15\ \mathrm{pF}$
- $\rightarrow C'_{ox} = 11.5\ \mathrm{fF/(\mu m)^2}$

$$C'_{ox} = \frac{\epsilon_{ox}\epsilon_0}{t_{ox}} = \underbrace{12\ \mathrm{fF/(\mu m^2)}}_{\mathrm{typ}}$$

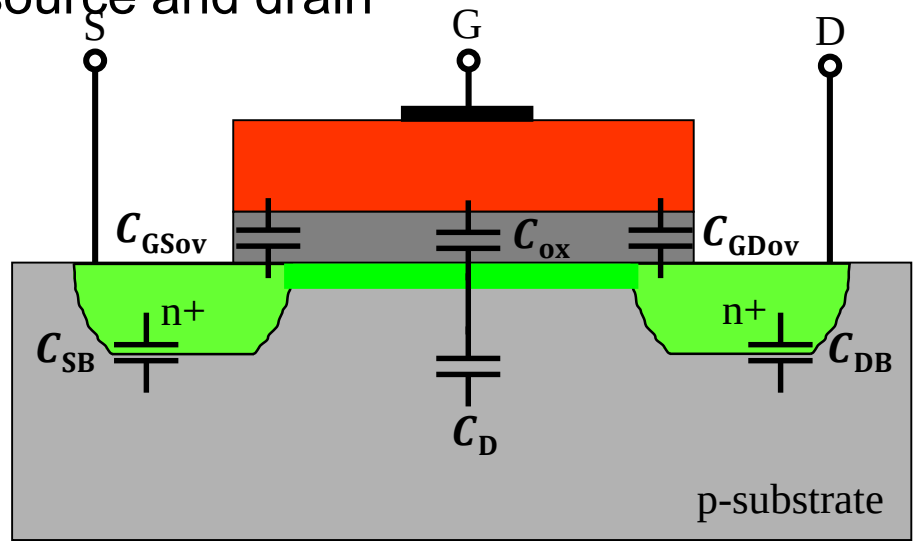


Capacitances – Dependence on Operating Point

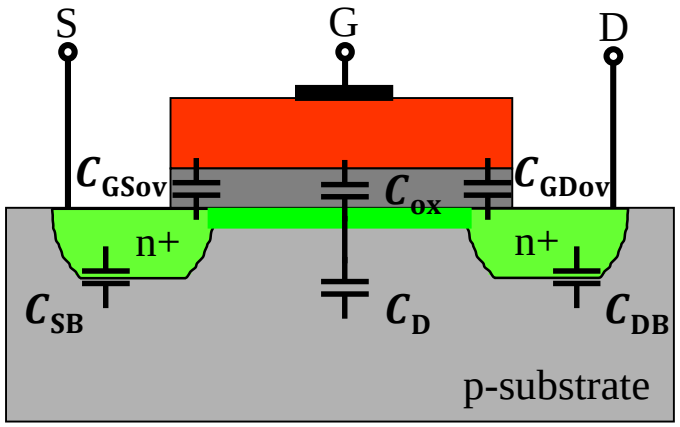
- Subthreshold: $v_{\mathrm{GS}} < V_{\mathrm{th}}$
- no inversion layer → no channel between source and drain

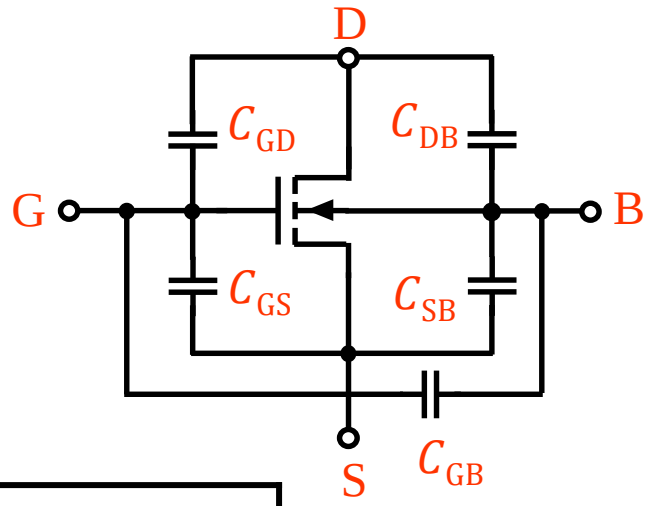
- Linear region: $v_{\mathrm{GS}} > V_{\mathrm{th}}$ and $v_{\mathrm{DS}} < v_{\mathrm{GS}} - V_{\mathrm{th}}$
- inversion layer → continuous channel between source and drain

- Saturation region: $v_{\mathrm{GS}} > V_{\mathrm{th}}$ and $v_{\mathrm{DS}} > v_{\mathrm{GS}} - V_{\mathrm{th}}$
- inversion layer → pinched-off channel between source and drain
- drain has “no connection” to channel



Capacitances – Linear Region





simulation: $v_{GS} = 0.5\ \text{V}, v_{DS} = 0.05\ \text{V}$
nMOS 0.13µm BSIM

same area

W/L = 10/10		W/L = 1/10		W/L = 10/1		W/L = 1/1	
Calc.	Sim.	Calc.	Sim.	Calc.	Sim.	Calc.	Sim.
1200f	1/10	120f		120f	1/10	12f	
	23f		3f		2.5f		0.3f
600f	455f	60f	45f	60f	43f	6f	4.3f
600f	570f	60f 1/10	56f	60f	60f	1/10 6f	5.99f
	5.3f		0.53f	~W~const	5.3f		0.53f
	5.4f	1/10	0.54f		5.4f	1/10	0.54

C_{ox}

C_{GB}

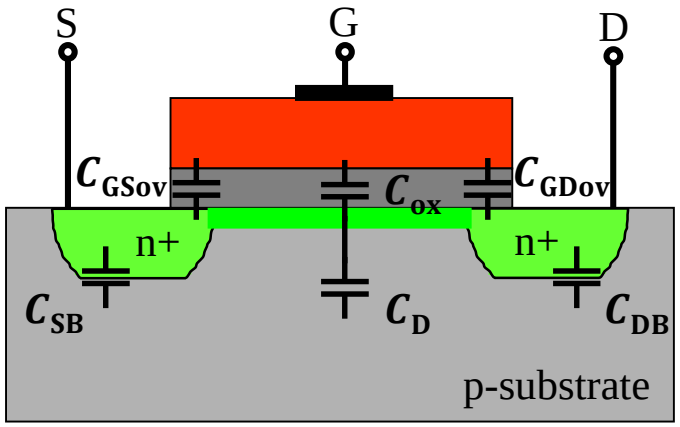
$C_{GD} \approx 0.5 \cdot C_{ox} + C_{GDov}$

$C_{GS} \approx 0.5 \cdot C_{ox} + C_{GSov}$

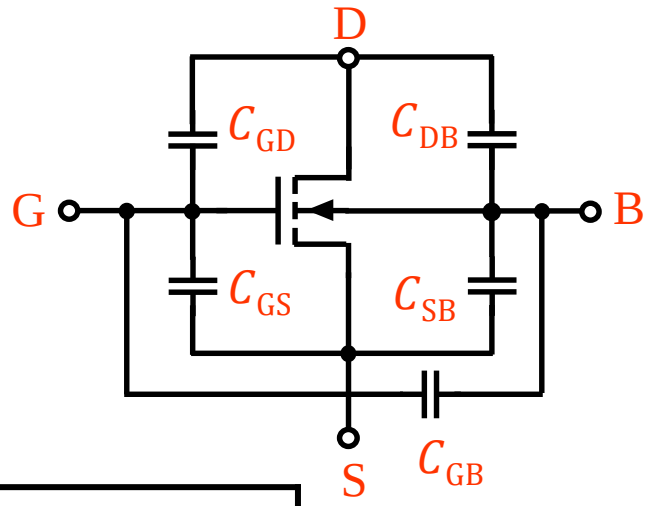
$C_{DB} \quad (L_{D,DIFF} = 0.75\mu\text{m})$

$C_{SB} \quad (L_{S,DIFF} = 0.75\mu\text{m})$

Capacitances – Saturation Region



simulation: $v_{GS} = 0.5\,V, v_{DS} = 1.2\,V$
nMOS 0.13µm BSIM



C_{ox}

C_{GB}

$C_{GD} \approx C_{GDov}$

$C_{GS} \approx 60\%\, C_{OX} + C_{GSov}$

C_{DB} $(L_{D,DIFF} = 0.75\mu m)$

C_{SB} $(L_{S,DIFF} = 0.75\mu m)$

W/L = 10/10		W/L = 1/10		W/L = 10/1		W/L = 1/1	
Calc.	Sim.	Calc.	Sim.	Calc.	Sim.	Calc.	Sim.
1200f		120f		120f		12f	
	50f		5f		4.5f		0.5f
	5f		0.5f		3f		0.3f
800f	747f	80f	74f	80f	78f	8f	7.8f
	4.1f		0.41f		4.1f		0.41f
	5.4f		0.54f		5.4f		0.54f

~W~const
~W~1/10
~W~const
1/10
1/10

MOS Transistor – Summary Equations

subthreshold region

$$v_{GS} < V_{th}$$

$$i_D \approx 0\ \mathrm{A}$$

valid for channels with *W*, *L* > 1 µm

linear region

$$v_{GS} > V_{th} \quad \&\& \quad v_{DS} < v_{GS} - V_{th}$$

$$i_D = \beta \left[(v_{GS} - V_{th}) v_{DS} - \frac{1}{2} v_{DS}^2 \right] (1 + \lambda v_{DS})$$

saturation region

$$v_{GS} > V_{th} \quad \&\& \quad v_{DS} > v_{GS} - V_{th}$$

$$i_D = \frac{\beta}{2} (v_{GS} - V_{th})^2 (1 + \lambda v_{DS})$$

with $\beta = \beta_0 \frac{W}{L} = \mu_n C'_{ox}$, $C'_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$

$$V_{th} = V_{th,0} + \gamma \left(\sqrt{2\Phi_F + v_{SB}} - \sqrt{2\Phi_F} \right)$$

Summary

- MOS transistor – second-order effects
- linear region
  - continuous channel between source and drain
  - $C_{\mathrm{GS}} \approx C_{\mathrm{GD}} \approx C_{\mathrm{ox}}/2$ (proportional to WL)
  - C_{DB} & C_{SB} are minimum one order of magnitude smaller than C_{GS} & C_{GD}
- saturation region
  - channel pinched off at drain → “connected” only to source
  - $C_{\mathrm{GS}} \approx 0.5 \ldots 0.75\ C_{\mathrm{ox}}$ (proportional to WL)
  - $C_{\mathrm{GD}} \approx C_{\mathrm{GDov}} \ll C_{\mathrm{GS}}$ → two orders of magnitude smaller (rough estimate), proportional to W
  - C_{DB} & C_{SB} are minimum one order of magnitude smaller than C_{GS}

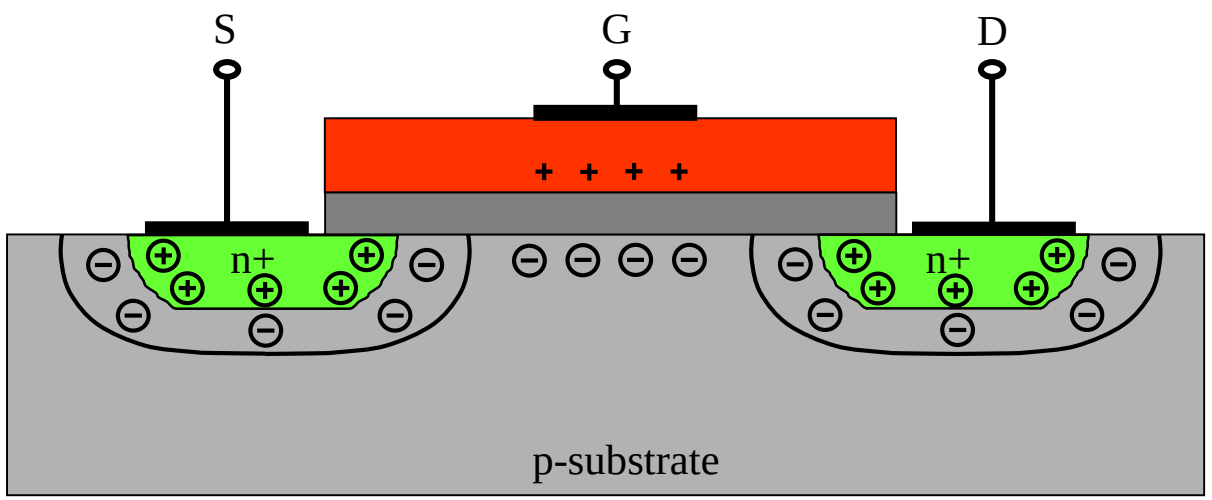
Literature

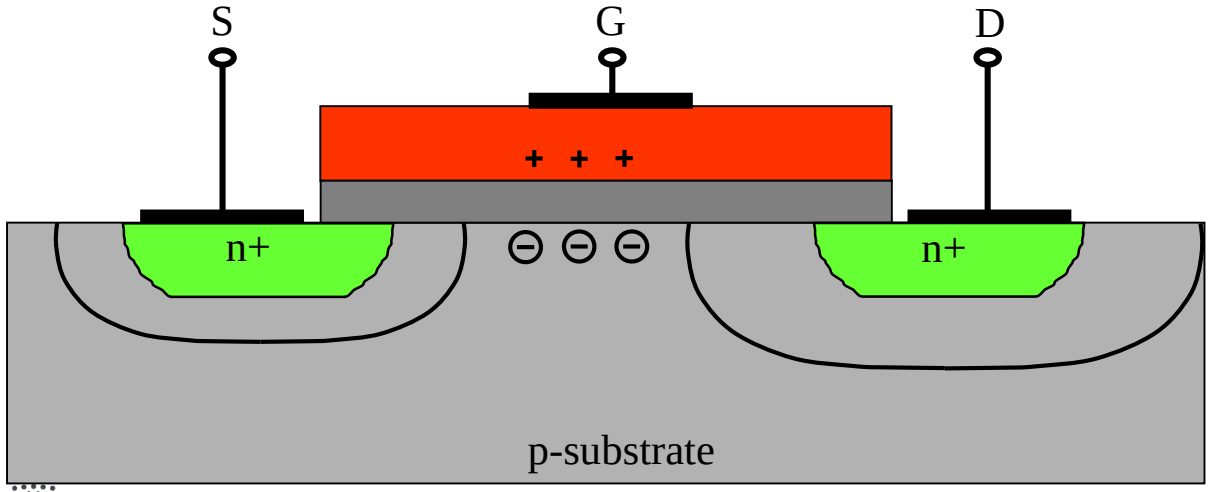
- Literature
- Razavi, chapters 2.3 & 2.4, pp. 23-38
- Sedra / Smith, chapter 4.8, pp. 320-325
- Allen / Holberg, chapter 3.3, pp. 87-92

Overview

- **MOS Transistor**
  - $i(v)$ – Characteristics
  - Second-order Effects
  - **Short-channel Effects**
    - **DB- & SB-Diodes & Drain Induced Barrier Lowering (DIBL)**
    - Mobility Degradation
    - Velocity Saturation
- Passive Devices & Wiring

DB/SB Diodes & DIBL





The region under the gate is not only depleted by the gate voltage but also partially depleted by the DB and SB pn-junctions!

Effect increases with shrinking device length!

$$\rightarrow \quad L \downarrow \quad \rightarrow \quad V_{\mathrm{th}} \downarrow$$

Effect is stronger when the transistor is operated in saturation (large V_{DS})!

→ drain induced barrier lowering (DIBL)

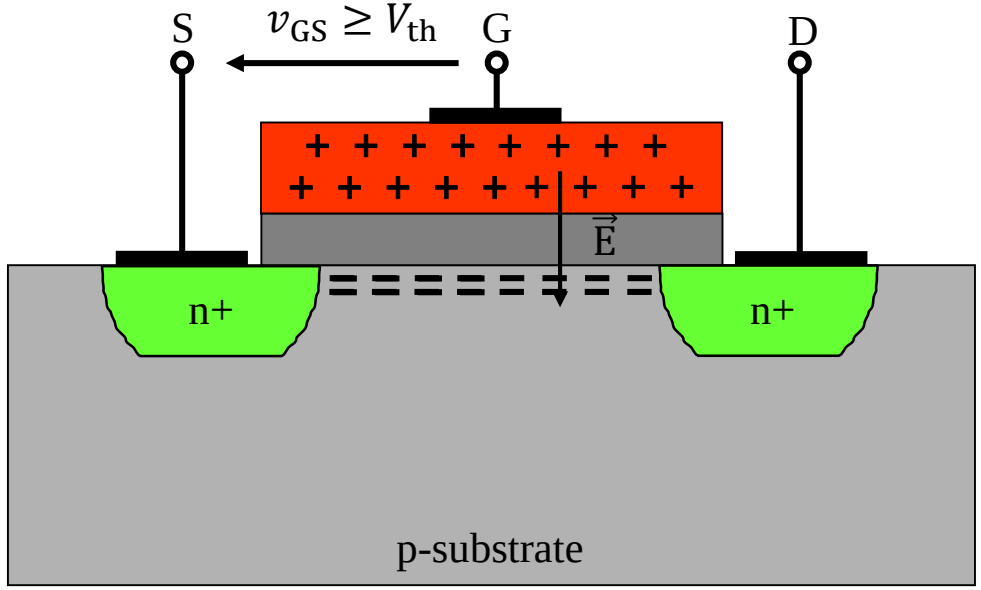
$$\rightarrow \quad v_{\mathrm{DS}} \uparrow \quad \rightarrow \quad V_{\mathrm{th}} \downarrow$$

$$\rightarrow \quad v_{\mathrm{th}} = f(L, v_{\mathrm{SB}}, v_{\mathrm{DS}})$$

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Mobility Degradation – Vertical Field



$\vec{E} \uparrow$

→ channel becomes “shallower”

→ carrier scattering

→ mobility $\mu \downarrow$

empirical model: $\mu_{eff} = \frac{\mu_0}{1 + \theta(v_{GS} - V_{th})}$ $\theta = ?$

saturation current: $i_D = \frac{1}{2} \frac{\mu_0 C'_{ox}}{[1 + \theta(v_{GS} - V_{th})]} \frac{W}{L} (v_{GS} - V_{th})^2$

Taylor-approximation for $\theta(v_{GS} - V_{th}) << 1$

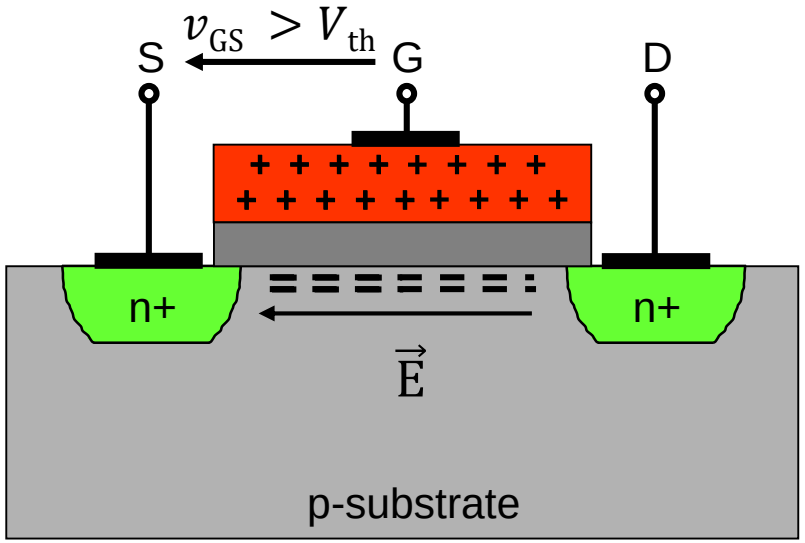
$$i_D \approx \frac{1}{2} \beta_0 \frac{W}{L} [(v_{GS} - V_{th})^2 - \theta(v_{GS} - V_{th})^3]$$

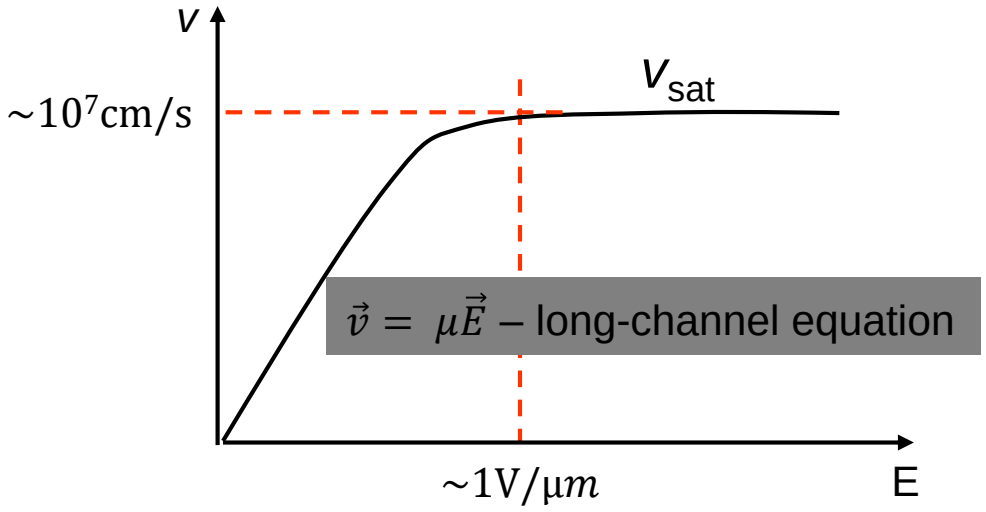
higher harmonics (third-order)

Overview

- **MOS Transistor**
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    - Mobility Degradation
    - **Velocity Saturation**
- Passive Devices & Interconnects

Velocity Saturation – Lateral Field

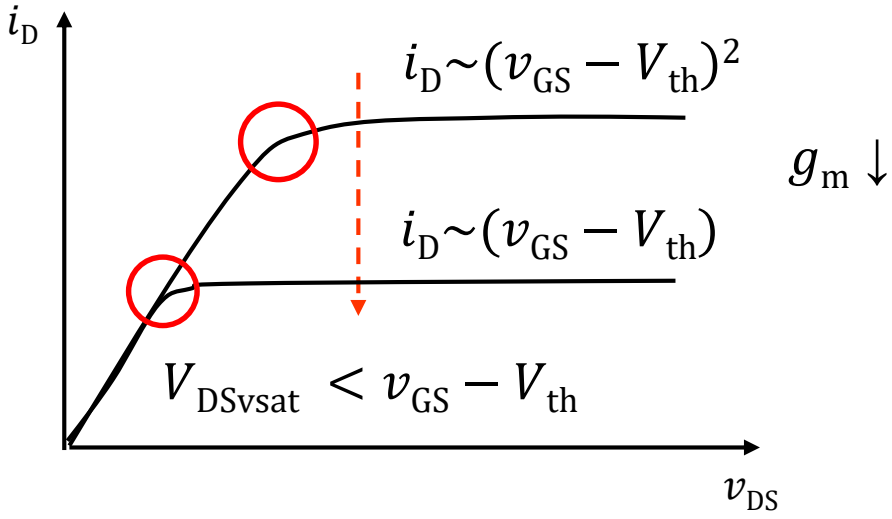




- saturation current

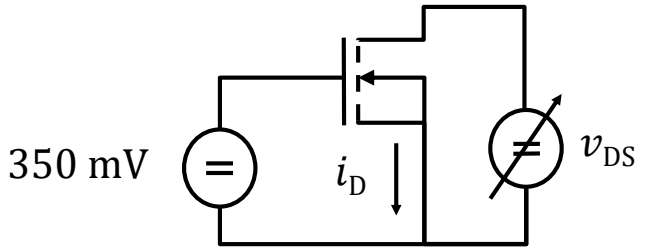
$$i_D = v_{sat} W C'_{ox} (v_{GS} - V_{th})$$

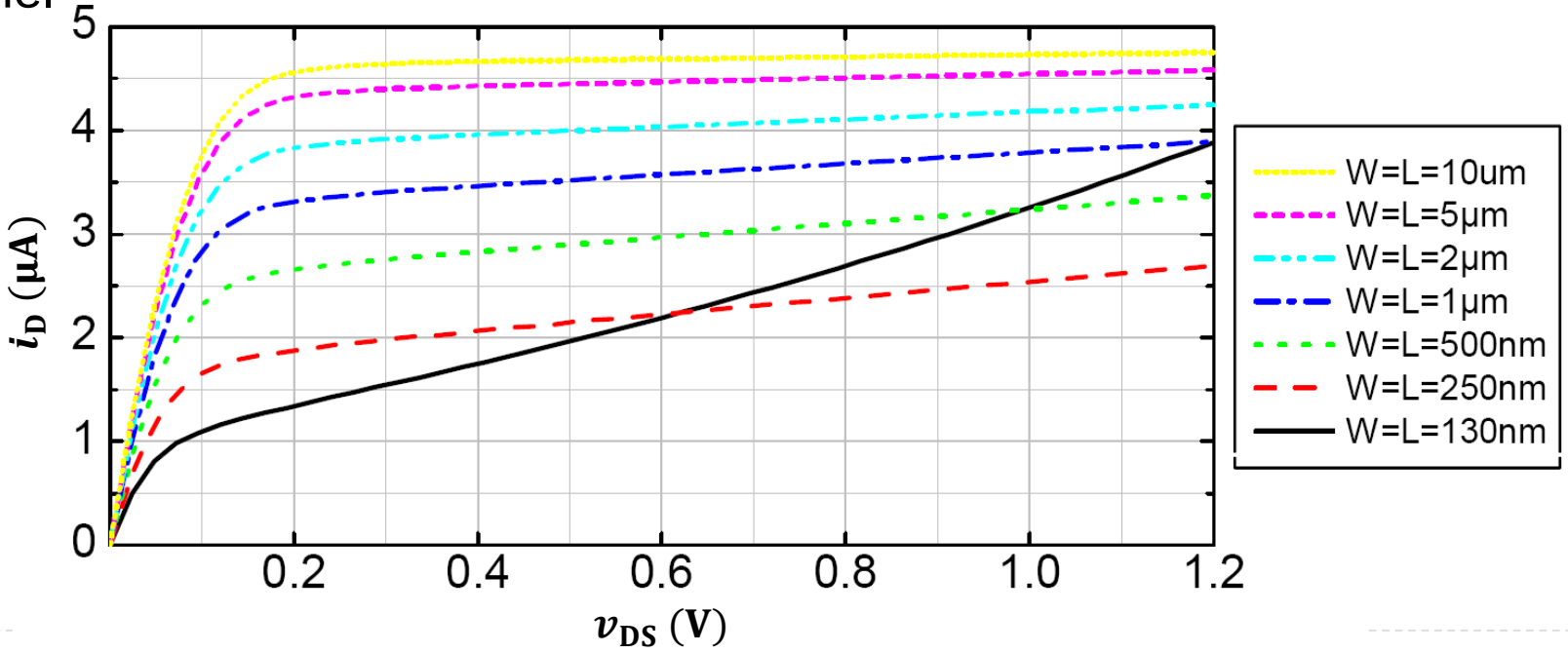
- i_D linearly and not quadratically proportional to $(v_{GS} - V_{th})$!



$i_{\mathrm{D}}(v_{\mathrm{DS}})$ - nMOS Transfer Characteristic

- All devices have the same W/L ratio.
- However, they drive different currents!
  --> short-channel effects
- long-channel parameters: $V_{\mathrm{thn}} = 0.22\ \mathrm{V},\ \beta_{0\mathrm{n}} = 475\ \mu\mathrm{A/V}^2$
- short-channel parameters: $V_{\mathrm{thn}} = xxx\ \mathrm{V},\ \beta_{0\mathrm{n}} = xxx\ \mu\mathrm{A/V}^2$





Transistor Models vs. Hand Calculations

- Level 1: long-channel / “square-law” model
  - easy-to-use / hand calculations
  - long outdated…
- Level 2: physical model
  - velocity saturation & threshold variation
- Level 3: half empirical model
  - curve fitting on measured instances
- **Level 4 (BSIM): empirical model**
  - **became standard in 1996**
  - **very accurate in modelling short-channel effects**
  - **not that accurate in modelling the behavior in weak & moderate inversion**
  - **inappropriate for hand calculations**
- EKV model: charge model
  - not that widespread
  - very accurate in modelling the behavior of long-channel MOS transistors in weak and moderate inversion
  - oversimplified formulation of short-channel effects

BSIM (Berkeley Short-channel IGFET Model)

- efficient, highly empirical model
- > hundreds of parameters in total
- DC model requires 100 parameters
- channel-length modulation
- subthreshold conduction
- geometric dependence of electrical parameters
- vertical field dependence of carrier mobility
- carrier velocity saturation
- drain-induced barrier lowering
- depletion charge sharing by source & drain
- non-uniform doping profile for ion-implanted devices

Microelectronics – Our Teaching Approach

- **hand calculations**: long-channel / „square law“ model
  - learn how large- & small-signal parameters scale with width, length, bias current, etc.
  - results match the order of magnitude very well, yet not the absolute value
    --> get absolute values & fine tune your circuits by means of simulations

- **simulations**: BSIM model
  - very accurate

- our concentration course ***Analog CMOS Circuit Design*** (every summer term)
  - learn the **g_m-over-I_D design approach**
    (look-up tables of pre-simulated parameters based on the BSIM technology file)
  - apply the **g_m-over-I_D design approach** to the design of circuits / hand calculations
  - simulate & fine tune your design using the BSIM technology file
  - result: excellent matching of hand calculations & simulations

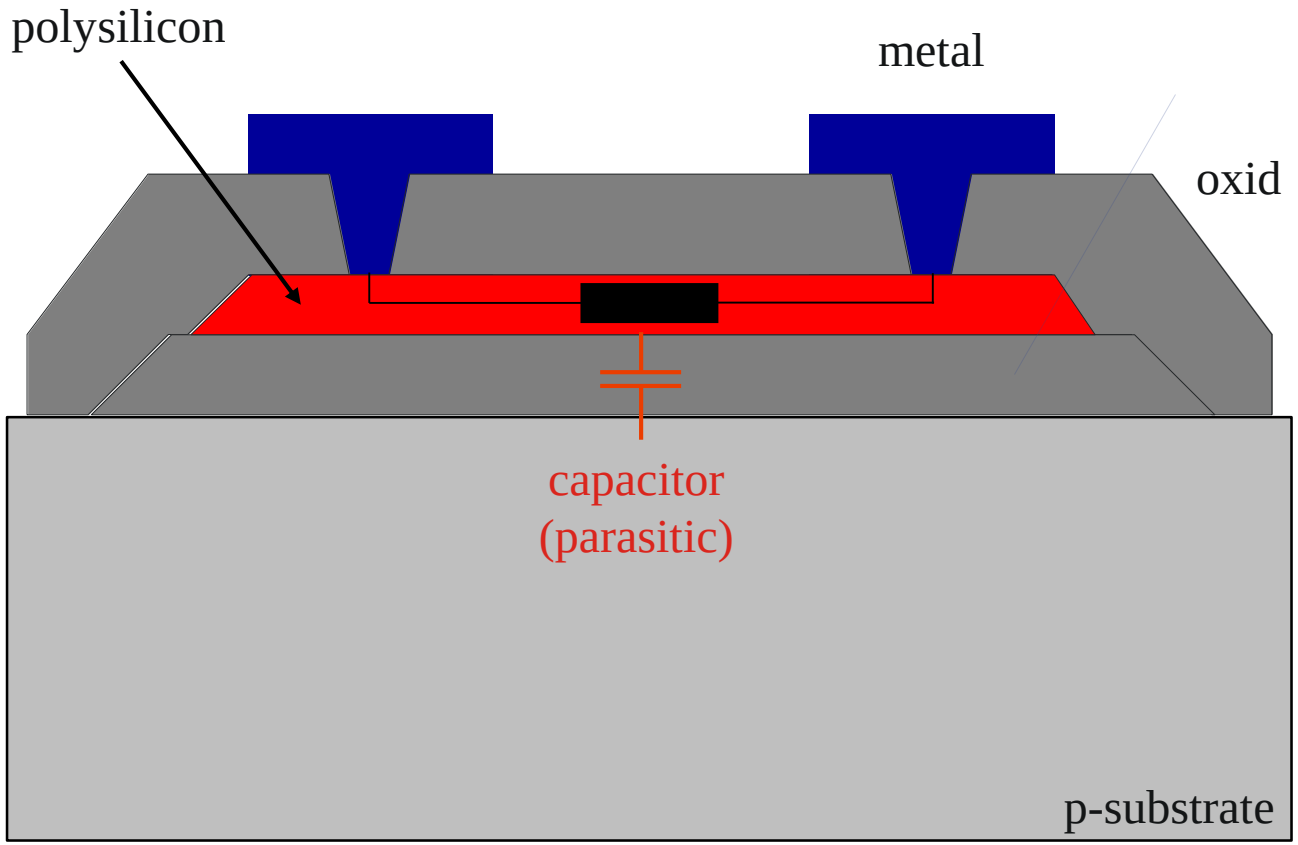
Literature

- Razavi, chapter 16, pp. 579-600

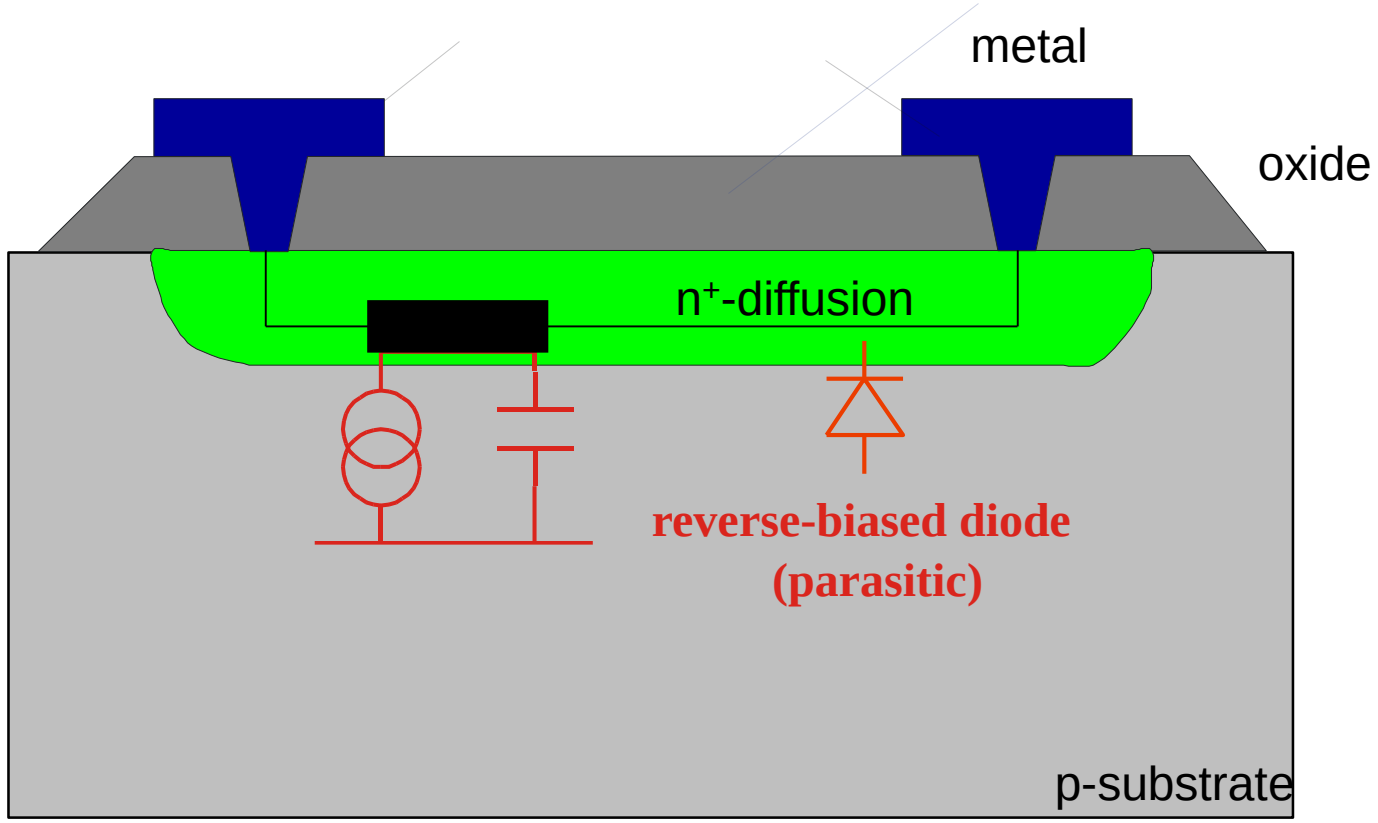
Overview

- MOS Transistor
  - $i(v)$ – Characteristics
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  - Capacitors
  - Wiring & Interconnects (Vias)

Poly Resistor

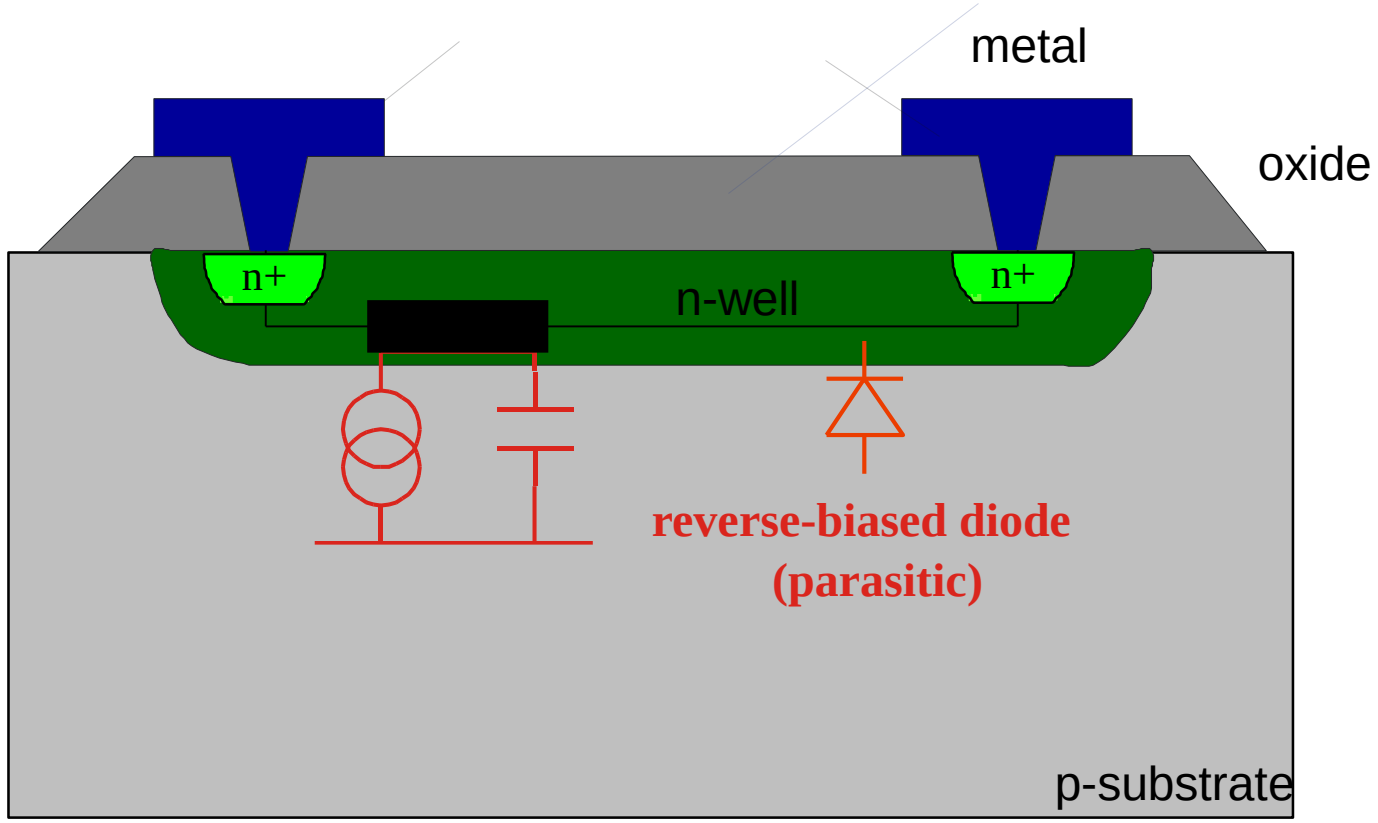


Diffusion Resistor

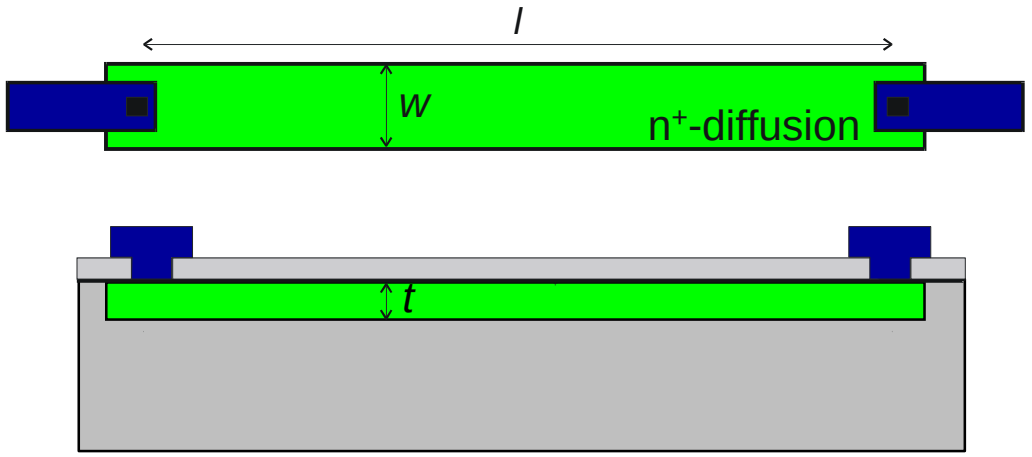


Typically, also p+-diffusion resistor available.

Well Resistor

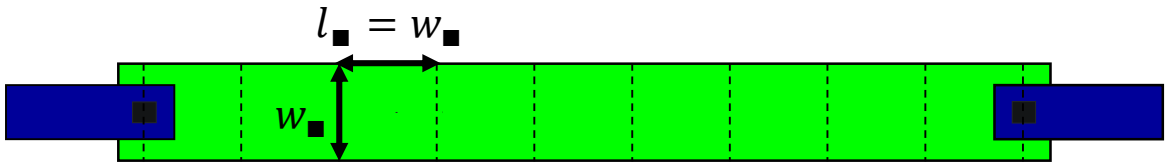


Total Resistance – Estimation



$$R = \rho \cdot \frac{l}{A} = \rho \cdot \frac{l}{w \cdot t} = \frac{\rho}{t} \cdot \frac{l}{w}$$

ρ – specific resistance of the material

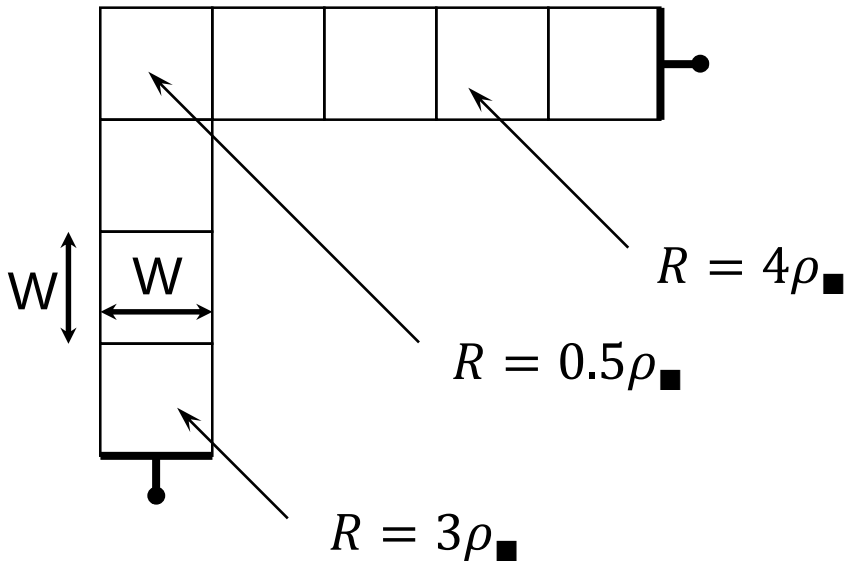


sheet/square resistance: $\rho_\blacksquare = \frac{\rho}{t}$

(refer to process documentation)

$$R = \rho_\blacksquare \frac{l}{w} = \rho_\blacksquare \, \#squares$$

Total Resistance – Estimation II



$$R = \rho_{\blacksquare}(0.5\#corners + \#squares)$$

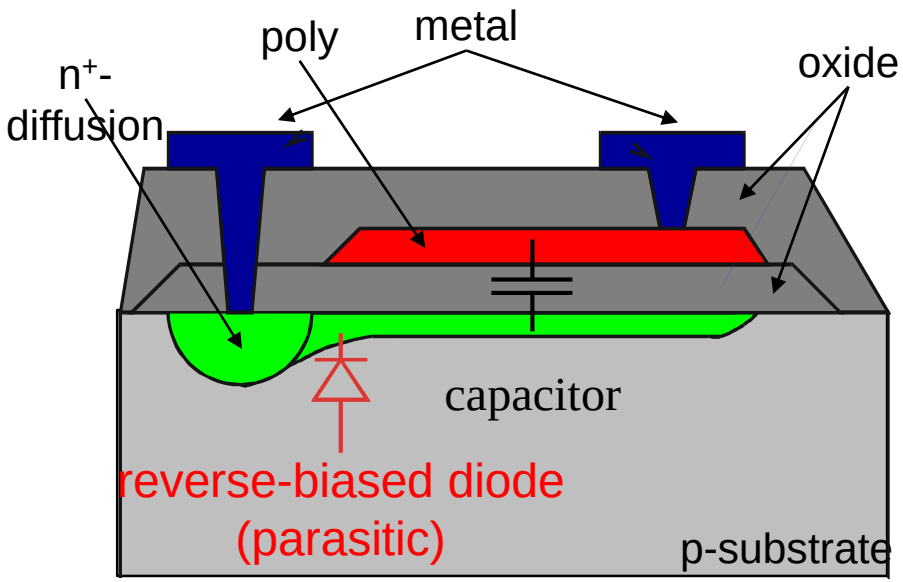
example: 0.13 µm technology

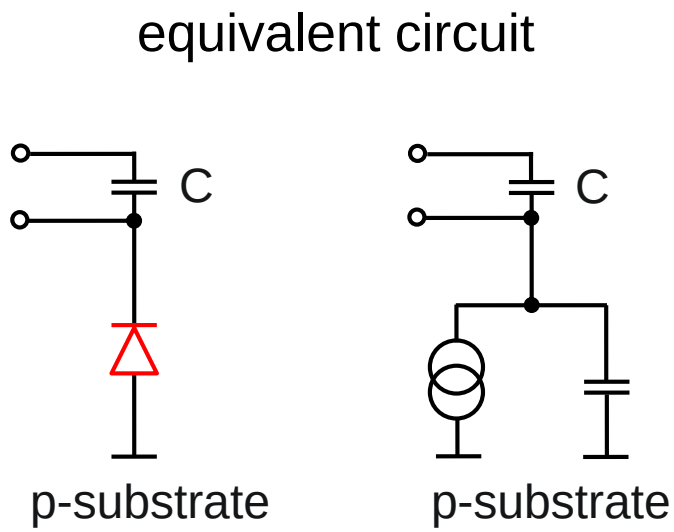
layer(s)	value			unit
	min	typ	max	
polysilicon	2.0	8.0	15.0	Ω/□
n+-diffusion				
p+-diffusion				
n-well				

Overview

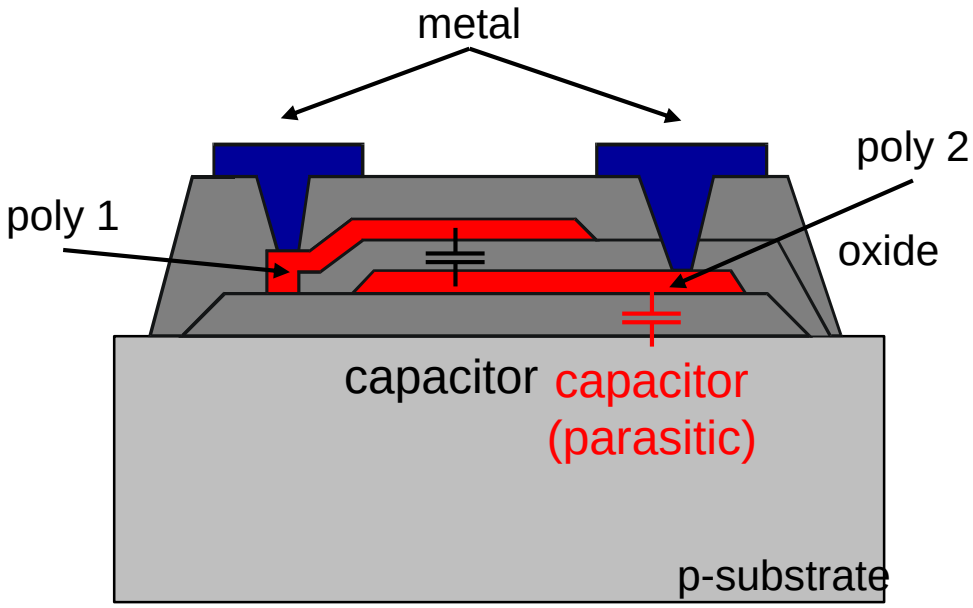
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Poly - n+ Capacitor

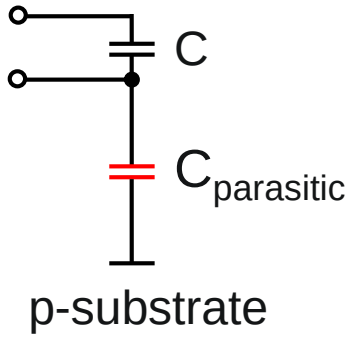




Poly - Poly Capacitor



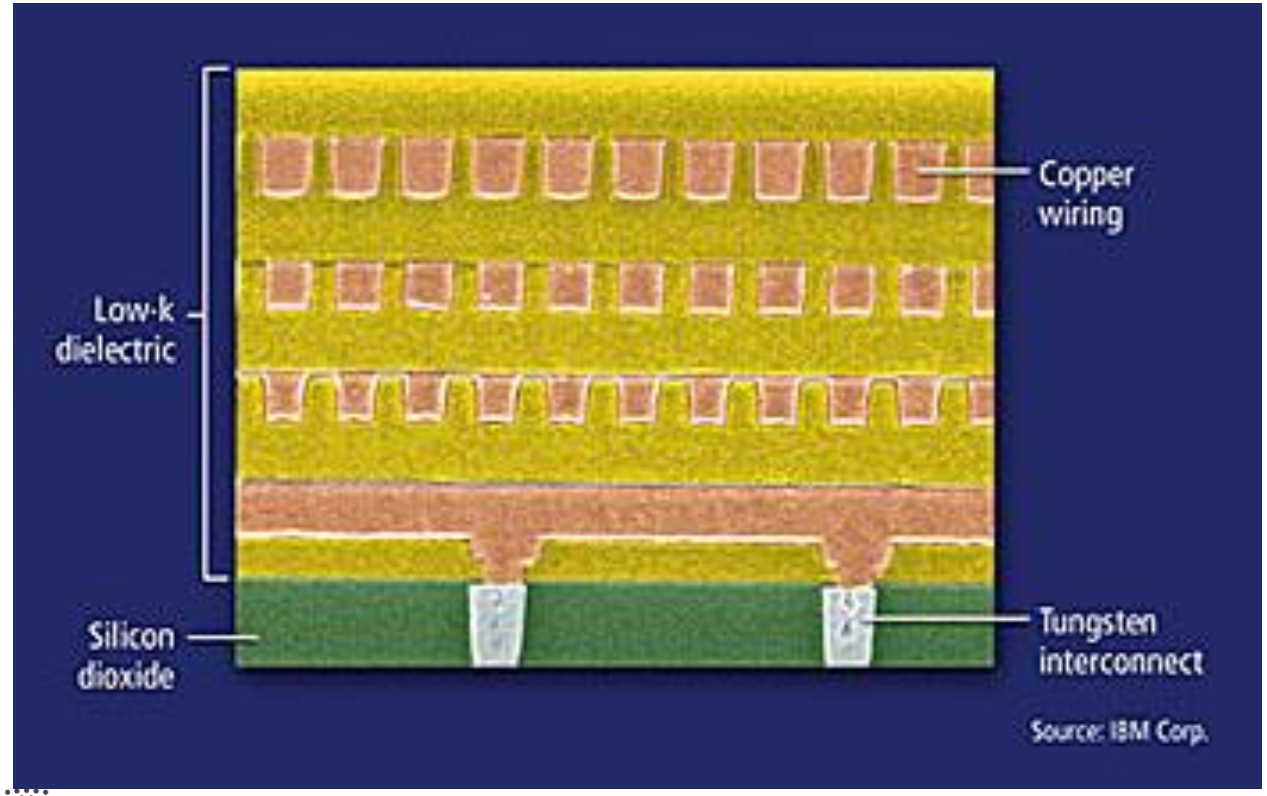
equivalent circuit

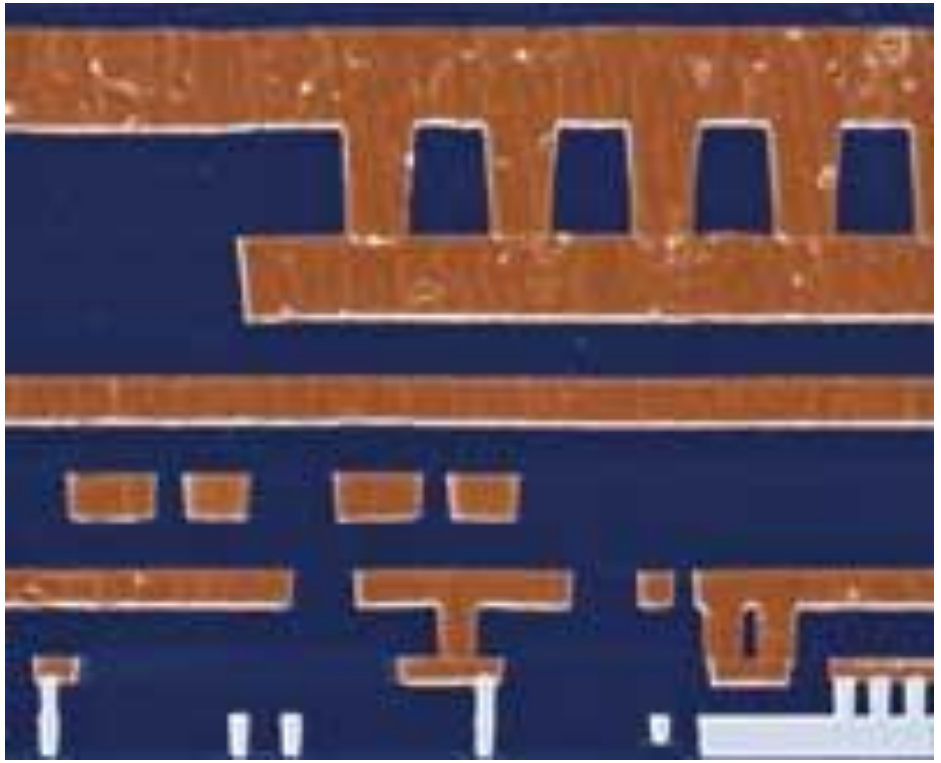


Overview

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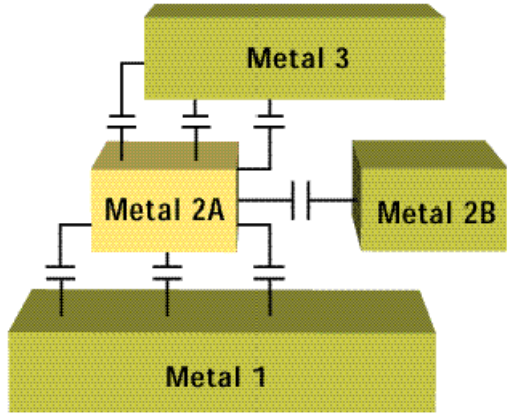
Metal Layers – Cross-sectional View





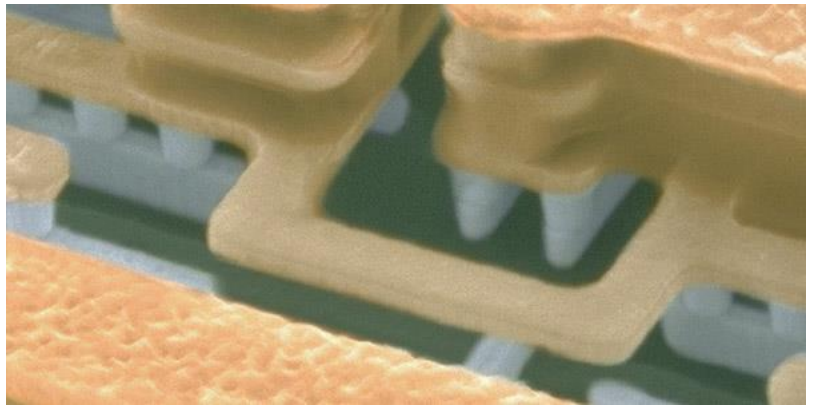
Metal Layers – 3D View





Metal Layers – Zoom

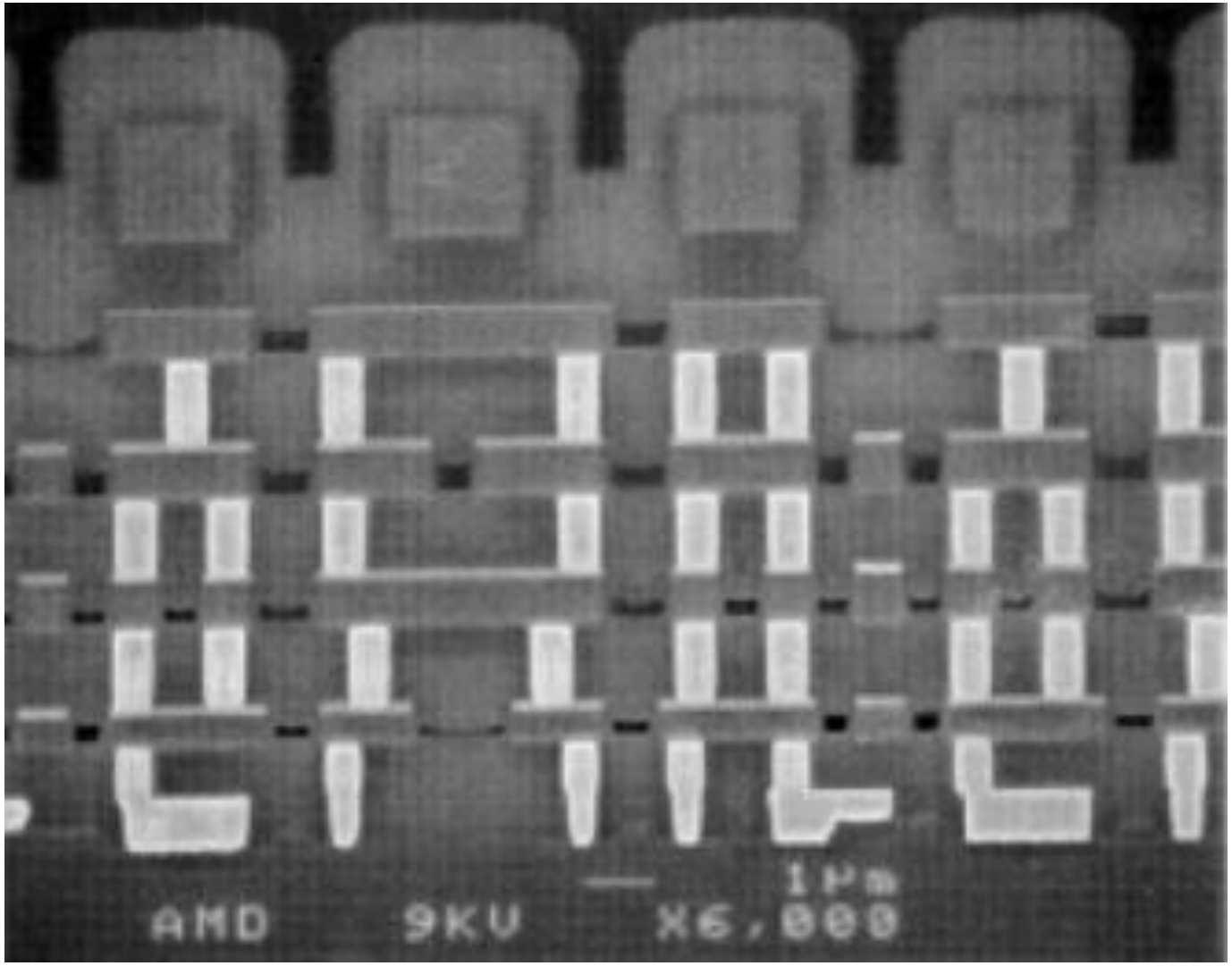




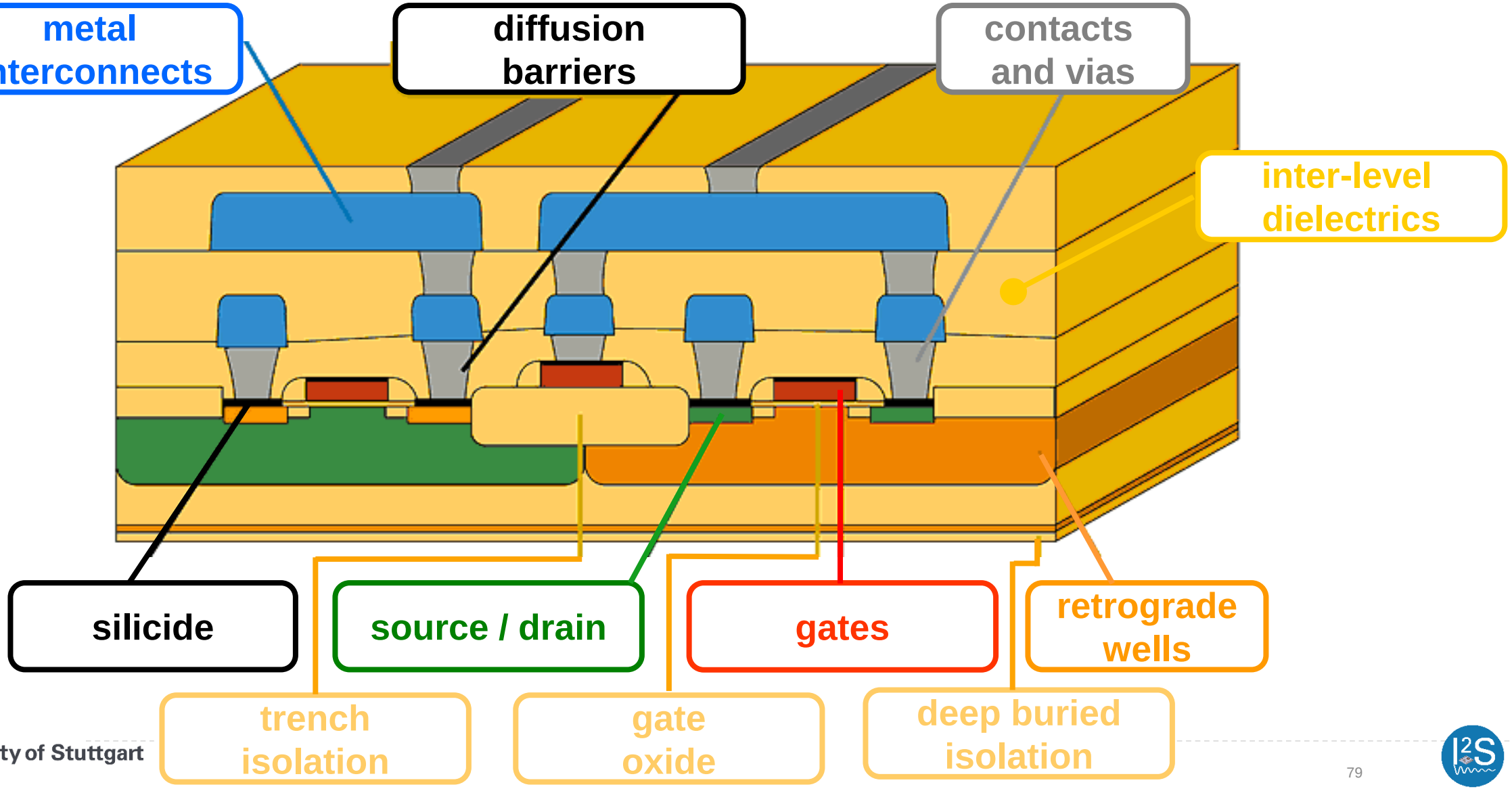
example: 0.13 µm technology

layer(s)	value			unit
	min	typ	max	
metal 7 – metal 8	19.0	27.0	35.0	mΩ/□
metal 1 – metal 6	49.0	70.0	91.0	mΩ/□
via 6 – via 7	0.1	0.6	3.0	Ω/via
via 1 – via 5	0.2	1.2	4.0	Ω/via
contact	2.0	10.0	25.0	Ω/cont

Cross-sectional View of an AMD Processor

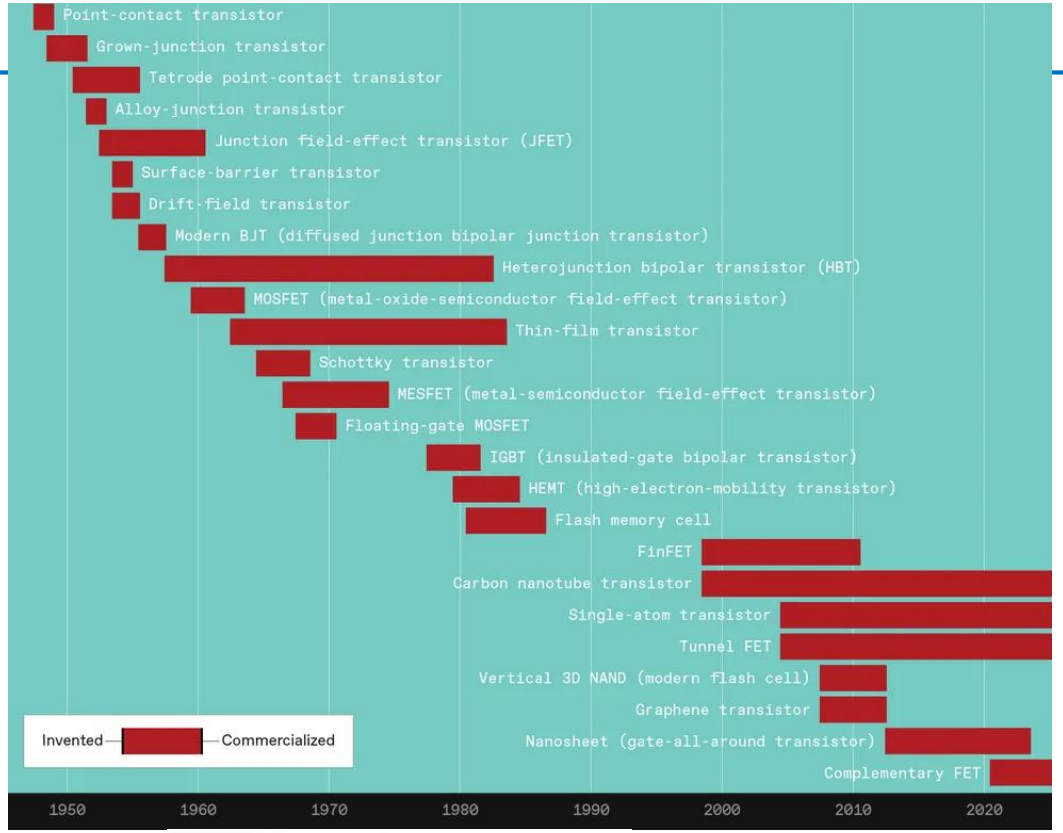


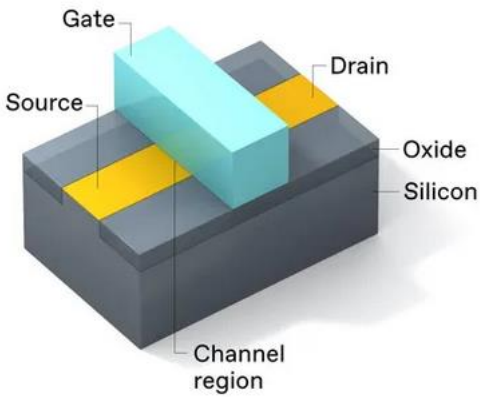
Classical CMOS Process – Cross-sectional View



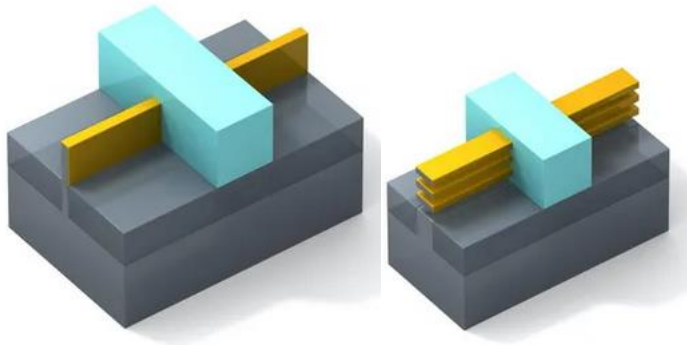


Source: IEEE Spectrum



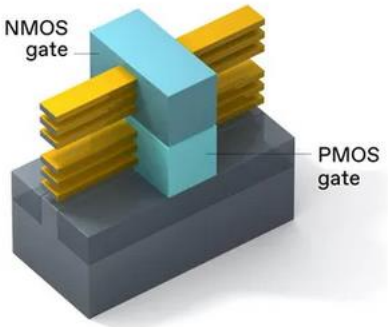


planar FET

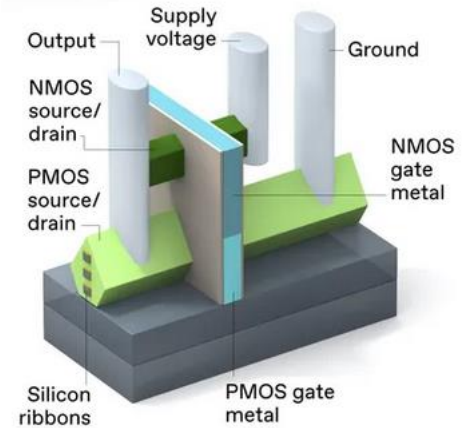


FinFET

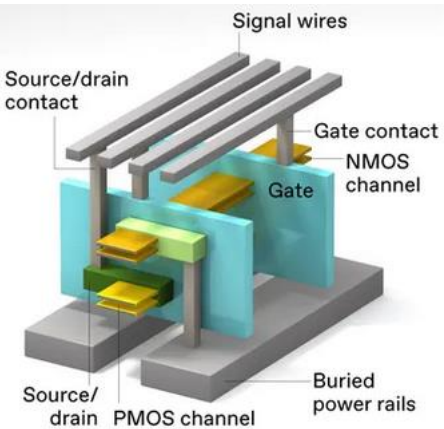
RibbonFET



stacked RibbonFET



stacked RibbonFET connections



stacked RibbonFET wireing